

The Practical Importance of Compositional Methods in Itonomic Data Analysis

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AI USE STATEMENT

I used Grammarly, a grammar and style-checking tool, to help review and improve the clarity and accuracy of the writing. I also used ChatGPT, an AI assistant developed by OpenAI, to help draft and refine portions of the R code used in the analyses and simulations after I specified the variables and goals. I then reviewed, revised, and refined the code using my own statistical knowledge. I take full responsibility for the final code, analyses, results, and interpretations in this report.

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Abstract

Ionic data consist of relative abundances of elemental components and are therefore inherently compositional. Despite this, they are often analyzed using standard methods applied directly to raw proportions, which may lead to misleading inference. This study evaluates the practical importance of treating ionic data as compositional in the context of a dataset of 113 spring amphipod samples from Oklahoma, each measured on 28 elemental components.

The study combines real-data analysis and simulation. In the real data, raw and centered log-ratio (CLR) representations produced different visual patterns, model behavior, and inferred relationships among elements. CLR generally revealed more structure and yielded more stable linear-model diagnostics than raw proportions. A simulation study comparing raw proportions, log-transformed raw proportions, and CLR showed that all three methods adequately controlled Type I error, but raw proportions had substantially lower power. Log-transformed proportions and CLR performed similarly in the main simulation. However, we also note, and demonstrate using an example, that the choice of log-based transformations may also depend on whether we wish to retain certain desirable properties such as sub-compositional coherence.

Overall, the findings suggest that raw proportions are generally inadequate for formal ionic inference. Log-based transformations improve statistical performance, while CLR is preferable when the objective is inference that respects the compositional structure of the data. When stable inference across subsets of elements is required, pairwise methods provide the strongest framework.

1. Introduction

1.1 Background and Motivation

Ionic data arise from the quantitative measurement of elemental composition within biological systems, capturing the relative abundance of multiple elements (such as calcium, magnesium, zinc, etc.) within a sample. Because these measurements represent parts of a whole, they are inherently compositional and constrained to sum to a constant (Aitchison, 1982; Pawlowsky-Glahn and Buccianti, 2011). As a result, ionic data do not lie in ordinary Euclidean space but in a simplex, where the components are interdependent.

Despite this structure, ionic data are often analyzed using conventional statistical methods applied directly to raw proportions. Such methods implicitly assume independence among variables and an unconstrained sample space, assumptions that are violated in compositional settings (Aitchison, 1986; Gloor et al., 2017). This mismatch can lead to misleading conclusions, including spurious associations and incorrect inference regarding the effects of biological or environmental predictors.

These concerns are not merely theoretical. Recent work in related fields has shown that ignoring the compositional nature of data can lead to substantial inferential errors. In microbiome research, for example, analyses based on raw proportions have been shown to produce misleading correlations and biased inference, whereas log-ratio-based methods yield more reliable and reproducible findings (Gloor et al., 2017; Weiss et al., 2017). Similar concerns are relevant to ionomics, where elemental abundances are likewise constrained and interdependent, but where the practical consequences of using compositional versus non-compositional approaches have not been extensively examined.

The motivating dataset for this study comes from an ecological survey of freshwater springs across Oklahoma, where amphipod samples were collected alongside environmental measurements to investigate whether spatial variation in amphipod traits is associated with elemental supply and other biotic or abiotic factors. Elemental composition was measured using ICP-OES, producing multivariate abundance profiles that are inherently compositional in nature. This study addresses the resulting methodological question by comparing raw and transformed approaches using both real data and simulation.

1.2 Statistical Challenges of Compositional Data

A defining feature of compositional data is the constant-sum constraint, which induces dependence among components. An increase in one component necessarily implies a decrease in at least one other, even if no true biological relationship exists. This creates several well-known statistical

difficulties, including spurious correlations, distorted variance structure, and subcompositional incoherence (Pearson, 1897; Aitchison, 1986; Pawlowsky-Glahn and Egozcue, 2006).

Traditional statistical methods, including linear regression, are developed for unconstrained data in Euclidean space. When these methods are applied directly to compositions, the resulting inference may not align with the relative structure of the data, especially when the scientific question concerns relationships among parts of a whole rather than independent absolute measurements.

1.3 Compositional Data Analysis

Compositional Data Analysis (CoDA), developed by Aitchison (1982, 1986), provides a principled framework for analyzing data in the simplex. Its central idea is to transform compositions into an unconstrained space using log-ratio transformations, thereby preserving relative information while accounting for the constant-sum constraint.

Among these transformations, the centered log-ratio (CLR) transformation is widely used. It expresses each component relative to the geometric mean of all components and maps the composition into real space, making it possible to apply standard multivariate methods while retaining a compositional interpretation.

Let $\mathbf{x} = (x_1, x_2, \dots, x_D)$ denote a composition of D positive components such that $x_i > 0$ for all i and $\sum_{i=1}^D x_i = 1$. The centered log-ratio (CLR) transformation expresses each component relative to the geometric mean of the full composition,

$$\text{clr}(\mathbf{x}) = \left(\log \frac{x_1}{g(\mathbf{x})}, \log \frac{x_2}{g(\mathbf{x})}, \dots, \log \frac{x_D}{g(\mathbf{x})} \right),$$

where

$$g(\mathbf{x}) = \left(\prod_{i=1}^D x_i \right)^{1/D}$$

is the geometric mean of the composition. This transformation maps the composition into real space, making it possible to apply standard multivariate methods while retaining a compositional interpretation.

CoDA using log-ratios has been widely used in fields such as geochemistry, microbiome research, and metabolomics, but its use in ionomics remains limited.

1.4 Research Objectives

The primary objective of this study is to evaluate the importance of treating ionomic data as compositional using log-ratio methods and to assess the consequences of ignoring this structure. Specifically, this study addresses four questions: how raw and CLR representations differ in visualization; how statistical inferences regarding sex, aquifer, and region differ under raw and CLR analyses; how raw proportions, log-transformed raw proportions, and CLR compare in terms of Type I error and power under controlled simulation.

1.5 Overview of the Study

To address these questions, the report first examines the real amphipod ionomic dataset using raw and CLR representations. It then evaluates the statistical performance of raw, LogRaw, and CLR methods in a simulation study designed to mimic the real dataset. It concludes by discussing the importance of desirable properties such as subcompositional coherence for ionomic inference.

1.6 Contribution of the Study

This study contributes to ionomics by providing a focused evaluation of how different analytical choices affect visualization, model behavior, statistical performance, and inferential interpretation in a compositional setting. It shows that the representation of ionomic data can meaningfully affect results, that some log-based transformation appears necessary for effective statistical inference, and that statistical performance alone is not sufficient when coherent interpretation across subsets of elements is required.

2. Literature Review

The statistical treatment of compositional data has been extensively developed since Aitchison's foundational work on the geometry of the simplex and the use of log-ratio transformations. Later work by Egozcue et al. (2003) and Pawlowsky-Glahn and Egozcue (2006) formalized the mathematical basis of Compositional Data Analysis (CoDA) and broadened its applicability across disciplines.

In recent years, compositional methods have gained broad acceptance in high-dimensional biological data analysis. In microbiome research, it is now widely recognized that sequencing data are inherently compositional and that failure to account for this structure can lead to biased inference and misleading associations (Gloor et al., 2017; Weiss et al., 2017). Similarly, in metabolomics and related omics fields, log-ratio transformations have been shown to improve validity and interpretability (Quinn et al., 2018). Recent work in glycomics further reinforces the practical importance of compositional thinking. Bennett et al. (2025) show that analyses based on raw relative abundances can generate spurious relationships and inflated false-positive rates, whereas CoDA methods provide more stable and reproducible inference.

At the same time, the literature does not support a completely uniform position always favoring CoDA using log-ratio transformations. Tomova et al. (2025) argue that the practical benefits of CoDA may depend on context, signal strength, and the inferential goal, while Greenacre (2023) cautions that commonly used transformations such as CLR are not always the most interpretable or necessary choice in practice. These perspectives suggest that the key question is not simply whether compositional methods improve performance, but under what conditions they matter and what kind of inference they support.

In ionomics, this debate remains largely unresolved. Although elemental abundances are clearly compositional, relatively little work has systematically evaluated how raw and compositional approaches differ in terms of visual interpretation, statistical performance, model behavior, and coherence of inference. This gap motivates the present study, which examines the practical importance of compositional methods in ionomics using both real data and controlled simulation.

3. Real Data Analysis

3.1 Data Description

The data used in this study were collected as part of a broader ecological survey of freshwater springs across Oklahoma, United States, aimed at investigating whether spatial variation in amphipod morphology is associated with environmental and geochemical factors. The focus here is on spring-dwelling amphipods and the relationship between elemental composition and biological or environmental predictors.

A total of 113 amphipod samples were collected from three regions of Oklahoma: Central Oklahoma (COK, $n = 35$), Northwest Oklahoma (NWOK, $n = 48$), and Southern Oklahoma (SOK, $n = 30$). These samples were obtained from six aquifer systems: Arbuckle–Simpson ($n = 30$), Cimarron River ($n = 18$), Garber–Wellington ($n = 12$), North Canadian River ($n = 12$), Ogallala ($n = 18$), and Rush Springs ($n = 23$). The dataset is nearly balanced with respect to sex, comprising 57 females and 56 males.

Elemental composition was measured for each sample, yielding concentrations ($\mu\text{g/g}$) for 28 elements: Al, As, B, Ba, Be, Bi, Ca, Cd, Co, Cr, Cu, Fe, K, Li, Mg, Mn, Mo, Na, Ni, P, Pb, S, Se, Si, Sr, Tl, V, and Zn. Amphipod samples were dried and analyzed using inductively coupled plasma optical emission spectroscopy (ICP-OES). For the purposes of this study, the raw concentrations were converted to proportions so that the 28 components sum to one for each sample, yielding compositional data.

The primary predictors considered were sex, region, and aquifer. Site-level identifiers were excluded because of collinearity with region. Region and aquifer are also partially related, since some aquifers are geographically concentrated within particular regions, introducing potential confounding that should be considered when interpreting results. No samples were removed, and there were no missing values or zeros in the compositional variables.

3.2 Data Preparation and Analytical Framework

Elemental concentrations were converted to proportions by dividing each observation by the row sum of all 28 elemental measurements. This placed each sample in a compositional framework, where the data are constrained to the simplex.

To account for compositional structure, a centered log-ratio (CLR) transformation was applied to the full set of 28 elements for each sample using the `compositions` package in R. The CLR transformation expresses each component relative to the geometric mean of the composition and maps the data into real space.

Two analytical approaches were compared in the real-data analysis. In the first, linear models were fitted directly to raw proportions. In the second, the same models were fitted to CLR-transformed responses. For each of the 28 elements, a separate linear regression model was fitted with sex, aquifer, and region as predictors. Reduced models were obtained by removing one predictor at a time, and partial F-tests were performed using ANOVA. No multiple testing adjustment was applied in the real-data analysis; results are therefore interpreted comparatively.

3.2.1 Element-wise Regression Models

Let $i = 1, \dots, n$ index amphipod samples, where $n = 113$, and let $j = 1, \dots, D$ index elemental components, where $D = 28$. Let X_{ij} denote the observed concentration of element j in sample i . The corresponding raw proportion for element j in sample i is defined as

$$P_{ij} = \frac{X_{ij}}{\sum_{k=1}^D X_{ik}}.$$

Thus, P_{ij} represents the proportion of the total elemental abundance in sample i contributed by element j .

For the CLR-based analysis, let

$$Z_{ij} = \log \left(\frac{P_{ij}}{g(\mathbf{P}_i)} \right),$$

where

$$g(\mathbf{P}_i) = \left(\prod_{k=1}^D P_{ik} \right)^{1/D}$$

is the geometric mean of the 28-part composition for sample i , and $\mathbf{P}_i = (P_{i1}, \dots, P_{iD})$.

For each element j , two separate linear regression models were fitted: one using the raw proportion P_{ij} as the response and one using the CLR-transformed value Z_{ij} as the response. The raw model was

$$P_{ij} = \beta_{0j}^{(R)} + \beta_{1j}^{(R)} \text{Male}_i + \sum_{a=1}^5 \gamma_{aj}^{(R)} A_{ia} + \sum_{r=1}^2 \delta_{rj}^{(R)} R_{ir} + \varepsilon_{ij}^{(R)},$$

and the CLR model was

$$Z_{ij} = \beta_{0j}^{(C)} + \beta_{1j}^{(C)} \text{Male}_i + \sum_{a=1}^5 \gamma_{aj}^{(C)} A_{ia} + \sum_{r=1}^2 \delta_{rj}^{(C)} R_{ir} + \varepsilon_{ij}^{(C)}.$$

Here, Male_i is an indicator variable equal to 1 if sample i is male and 0 if female; A_{ia} are aquifer indicator variables for the non-reference aquifer levels (Arbuckle–Simpson is the reference group); and R_{ir} are region indicator variables for the non-reference region levels (COK being the reference group). The intercepts $\beta_{0j}^{(R)}$ and $\beta_{0j}^{(C)}$ represent the expected raw proportion and CLR value, respectively, for the reference group. The coefficients $\beta_{1j}^{(R)}$ and $\beta_{1j}^{(C)}$ represent the sex effect for element j , while $\gamma_{aj}^{(R)}, \gamma_{aj}^{(C)}$ and $\delta_{rj}^{(R)}, \delta_{rj}^{(C)}$ represent aquifer and region effects. The error terms are denoted by $\varepsilon_{ij}^{(R)}$ and $\varepsilon_{ij}^{(C)}$. Error terms are assumed to be independently and identically Normally distributed with zero mean and constant variance.

Reduced models were obtained by removing one predictor at a time from the full model, and partial F-tests were performed using ANOVA to assess the contribution of sex, aquifer, and region for each element.

Interpretation of Some Estimates

For calcium in the raw-proportion model, the estimated coefficient for sex was

$$\hat{\beta}_{1,\text{Ca}}^{(R)} = 0.0217$$

This means that, after adjusting for aquifer and region, male amphipod samples were estimated to have calcium proportions that were on average 0.0217 higher than those of female samples.

For calcium in the CLR model, the corresponding sex coefficient was

$$\hat{\beta}_{1,\text{Ca}}^{(C)} = 0.1487$$

This estimate is interpreted differently. Because the CLR expresses calcium relative to the geometric mean of the full composition, a positive coefficient indicates that calcium tends to be relatively more abundant, compared with the rest of the ionome, in male samples than in female samples after accounting for aquifer and region.

As a second illustration, for zinc in the raw-proportion model, the estimated sex effect was

$$\hat{\beta}_{1,\text{Zn}}^{(R)} = -3.965 \times 10^{-5}$$

This means that, after adjusting for aquifer and region, male amphipod samples were estimated to have zinc proportions that were on average 3.965×10^{-5} lower than those of female samples. In contrast, the zinc CLR model gave

$$\hat{\beta}_{1,\text{Zn}}^{(C)} = -0.00424,$$

this estimate indicates that zinc tends to be very slightly lower, relative to the rest of the ionome, in male samples than in female samples after accounting for aquifer and region.

3.3 Visualization of Data Structure

To compare raw and compositional representations of the ionome, graphical summaries were constructed using both raw proportions and CLR-transformed values. These included bar plots of mean compositions across groups and principal component analysis (PCA) of the multivariate elemental profiles.

Raw bar plots by aquifer and sex show that one element dominates the composition in every group, with the remaining elements compressed into narrow segments of the stacked bars (Figures 3.1 and 3.2). This makes it difficult to compare moderate and low-abundance elements across groups. At the same time, the raw plots provide a clear display of the dominance structure of the ionome and are useful for identifying major contributors to total composition.

The CLR bar plots provide a more balanced representation (Figures 3.3 and 3.4). By expressing each component relative to the geometric mean, and using the logarithm to shrink the extreme cases, the CLR transformation spreads the elements across positive and negative values, making moderate and low-abundance elements much easier to compare. Differences among aquifers appear more visible in the CLR representation than in the raw plots, particularly for elements that are difficult to distinguish on the proportion scale.

Principal component analysis reveals similar differences between the two representations. In the raw PCA, variation is concentrated largely along the first principal component, producing a dominant one-dimensional pattern. In the CLR PCA, variation is spread more evenly across the first two components, yielding a more two-dimensional representation that reveals additional structure (Figures 3.5 and 3.6). The loading geometry also changes between raw and CLR PCA. Thus, the transformation alters not only scale but also the apparent relationships among components. Neither approach produces complete separation of aquifer groups, and no meaningful clustering by sex is observed (**Appendix Figure A.1 and A.2**).

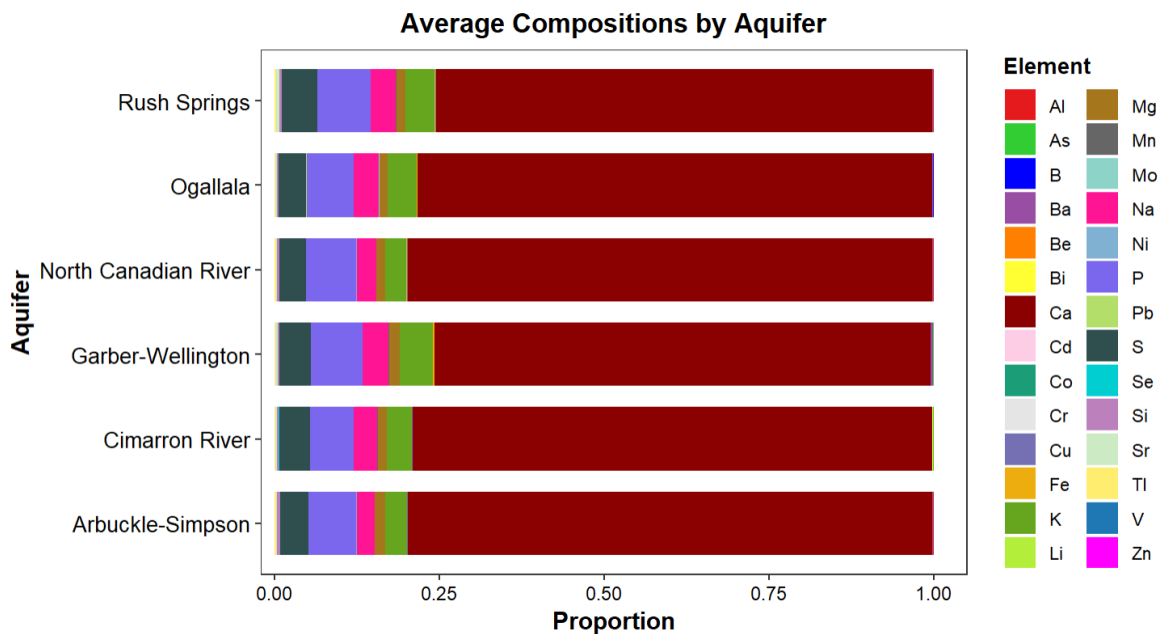


Figure 3.1. Average raw elemental compositions by aquifer. Stacked bar plots of mean raw proportions for the 28 elements across aquifer groups. The plot emphasizes dominance structure but compresses variation among lower-abundance elements.

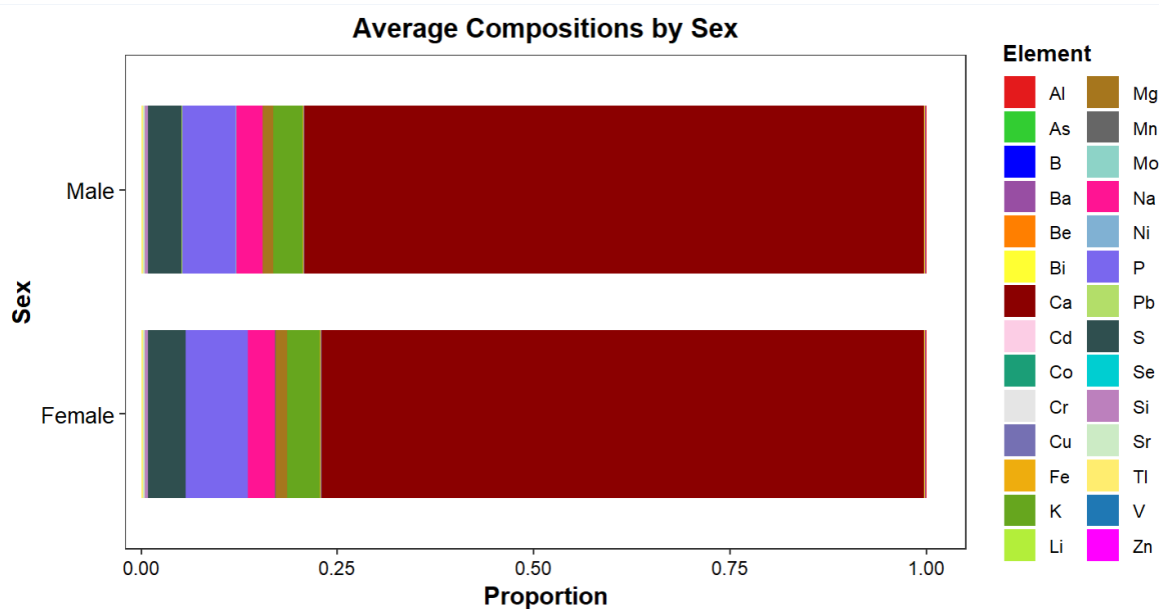


Figure 3.2. Average raw elemental compositions by sex. Stacked bar plots of mean raw proportions for the 28 elements across sex groups. The similarity in overall profiles illustrates the limited visual separation of groups on the raw scale.

Mean CLR-Transformed Ionome of Pods by Aquifer

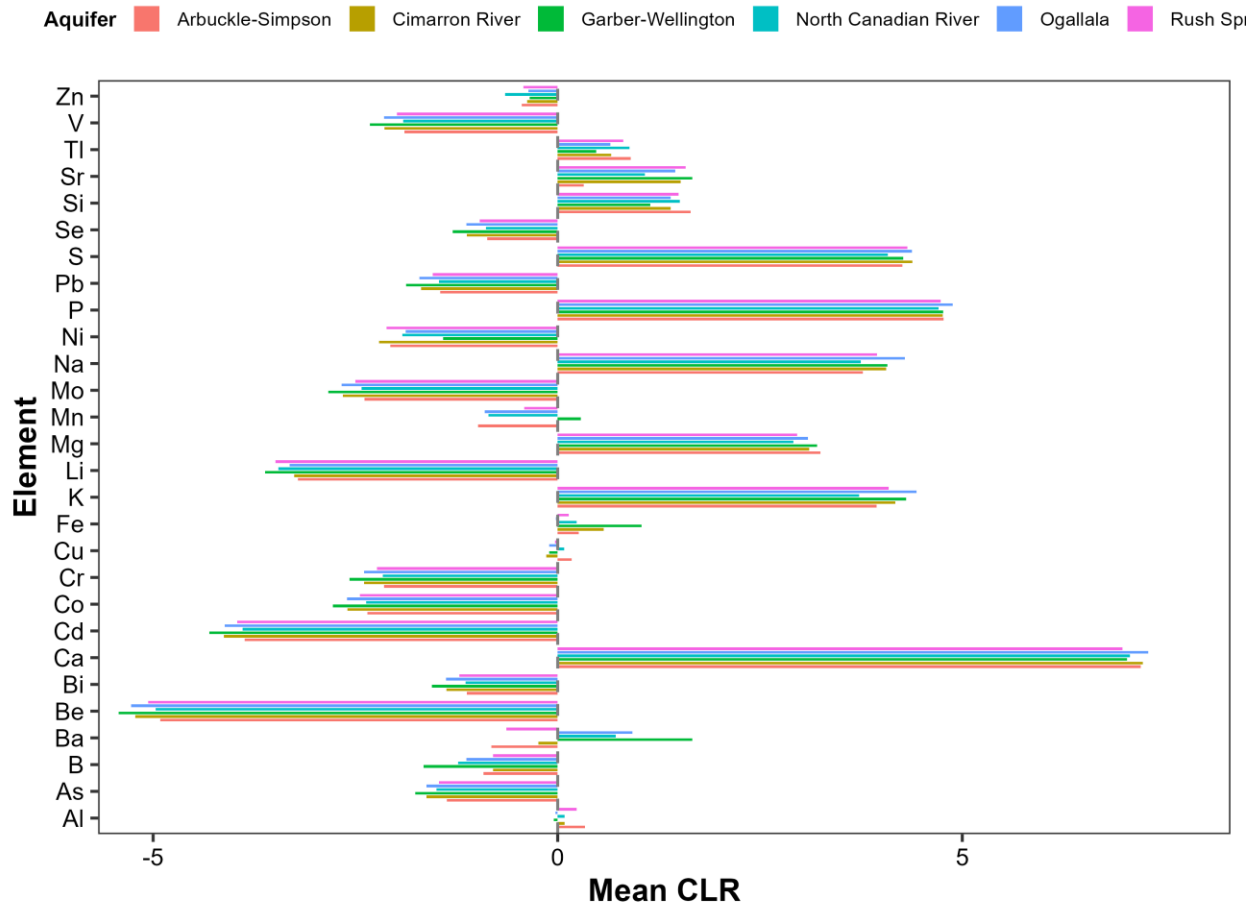


Figure 3.3. Average CLR-transformed elemental compositions by aquifer. Grouped bar plots of mean centered log-ratio (CLR) values across aquifer groups. Positive values indicate elements above the geometric mean and negative values indicate elements below it.

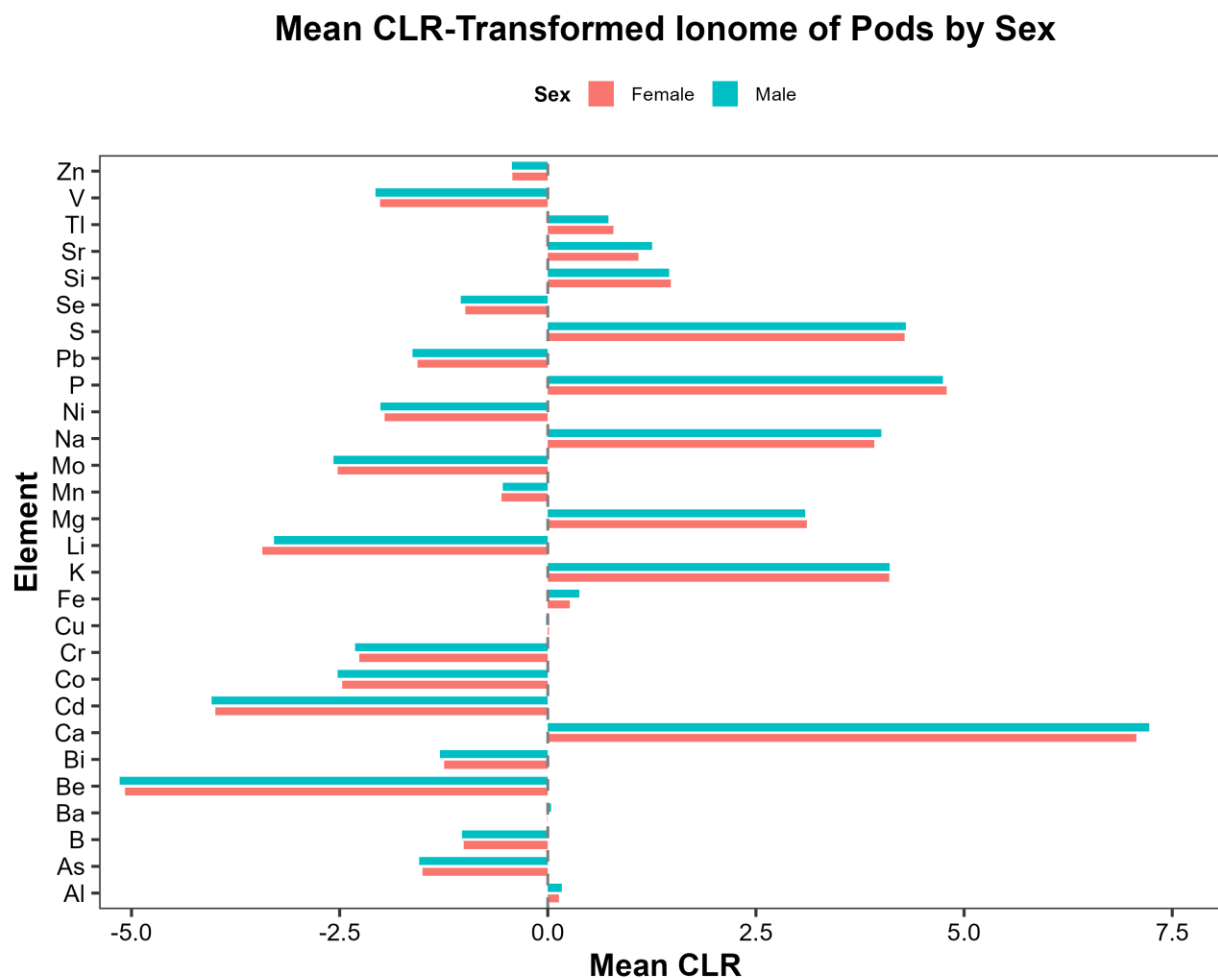


Figure 3.4. Average CLR-transformed elemental compositions by sex. Grouped bar plots of mean CLR values across sex groups. Compared with raw proportions, the CLR representation makes moderate and low-abundance elements easier to compare.

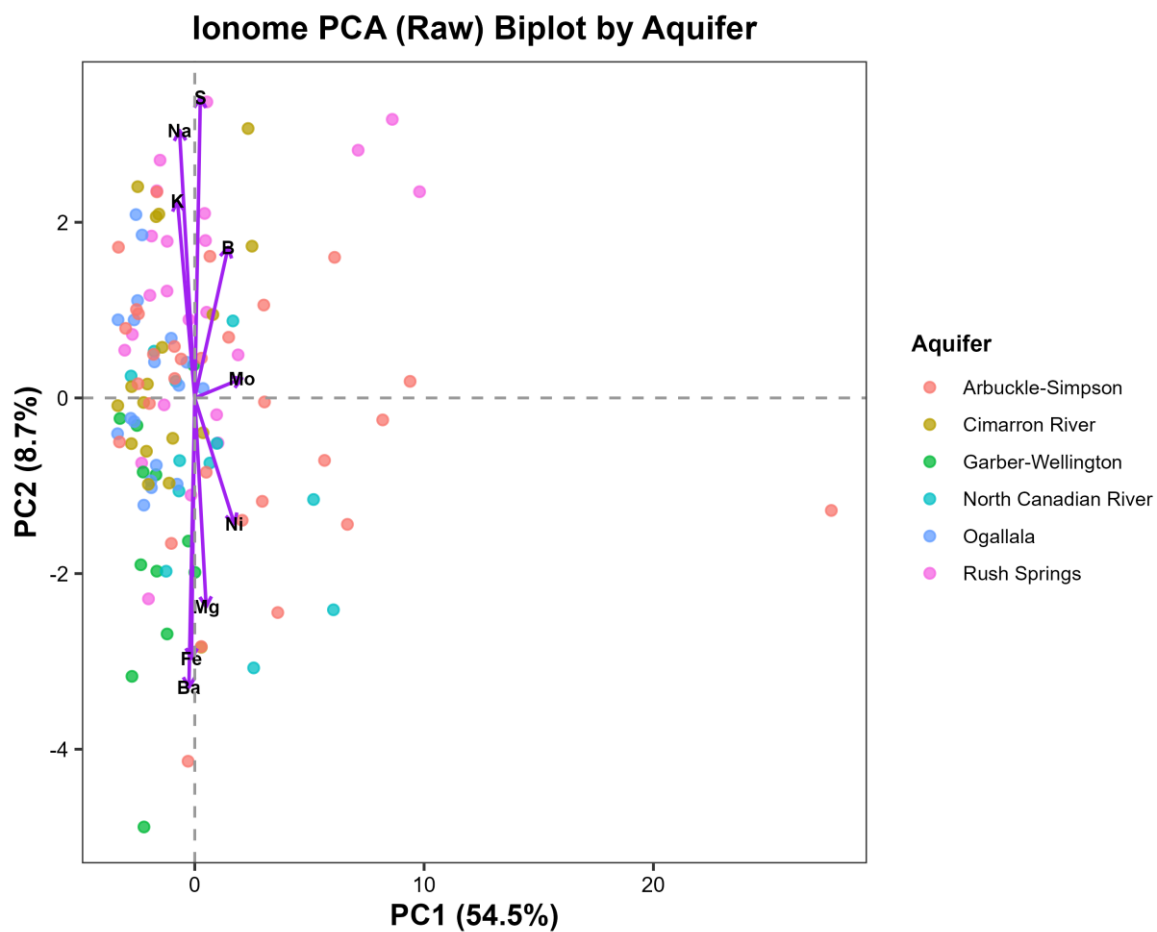


Figure 3.5. Principal component analysis of raw ionomic data by aquifer. Scores and biplot from PCA applied to centered and scaled raw elemental data. The raw representation concentrates much of the variation in the first principal component.

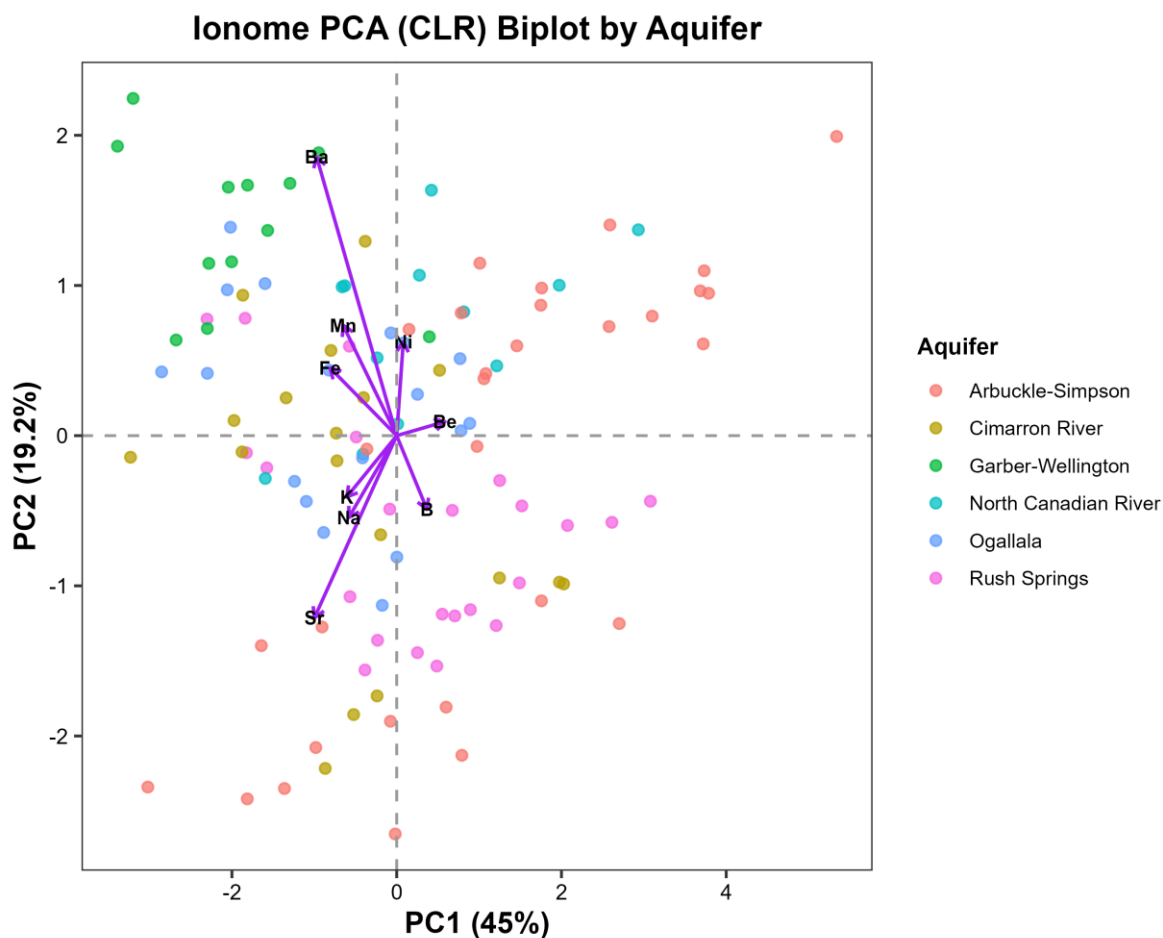


Figure 3.6. Principal component analysis of CLR-transformed ionomic data by aquifer. Scores and biplot from PCA applied to centered CLR-transformed data. The CLR representation spreads more evenly across components and reveals additional structure.

3.4 Comparison of Model Results

To compare raw and CLR-based inference, heatmaps summarizing significance patterns and coefficient directions were constructed across all 28 elements and predictors (Figures 3.7 and 3.8). Statistical significance in this section is based on unadjusted p-values.

Overall, the CLR-based analysis identifies more significant predictor–element relationships than the raw proportion approach. This suggests that CLR is more sensitive to subtle compositional changes, particularly those involving moderate and low-abundance elements. However, raw analysis still captures some strong associations, especially for dominant elements and stronger signals. Across both approaches, aquifer emerges as the strongest predictor, showing the most widespread associations with elemental composition.

The coefficient heatmaps (**Figure 3.8**) also show that some estimated effects change direction between raw and CLR representations. These sign changes are scientifically important because they show that the transformation affects not only which associations appear significant, but also how those effects are interpreted.

Graphical diagnostics were used to assess model assumptions for both raw and CLR-based regressions. The raw models frequently exhibit larger residual spread, departures from normality, non-constant variance, and greater sensitivity to influential observations, particularly for highly abundant elements (**Appendix Figure A.3 - A.5**). CLR generally improves these issues, producing more stable residual patterns and more consistent model behavior across elements, although it does not fully satisfy all assumptions in every case. A representative set of diagnostic plots is shown in Figure 3.9, with additional diagnostics provided in the appendix (**Appendix Figure A.6 - A.8**)

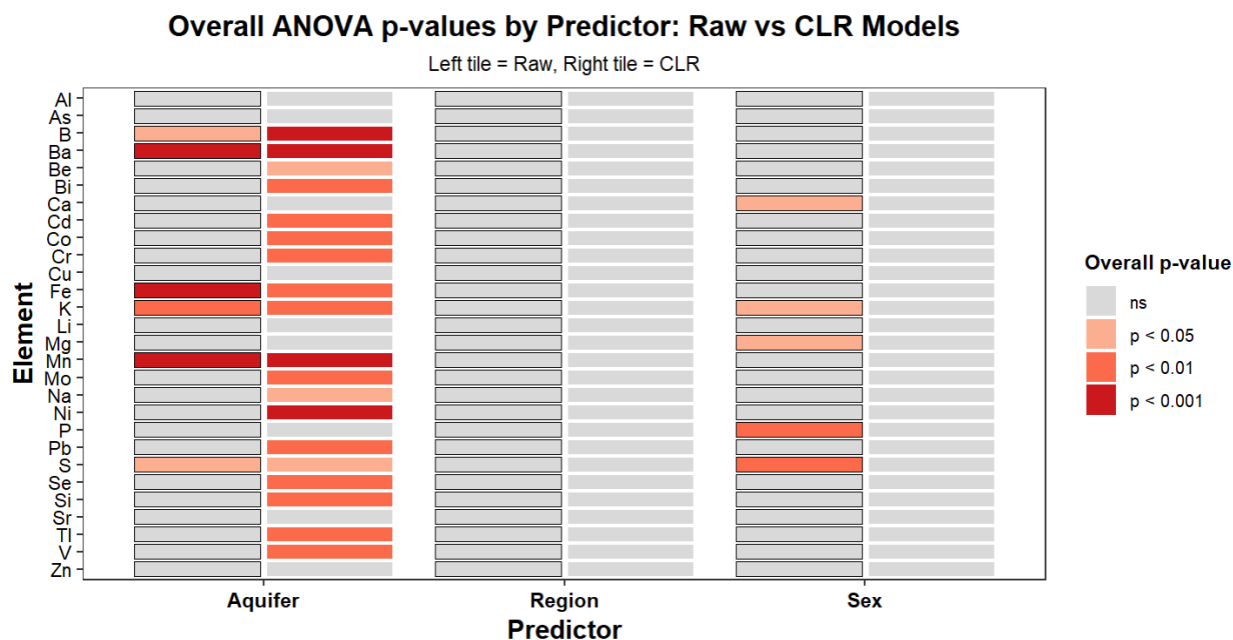


Figure 3.7. Heatmap of significance patterns under raw and CLR analyses. Heatmap summarizing significant predictor–element relationships across the 28 elements for raw and CLR-based regressions using unadjusted p-values.

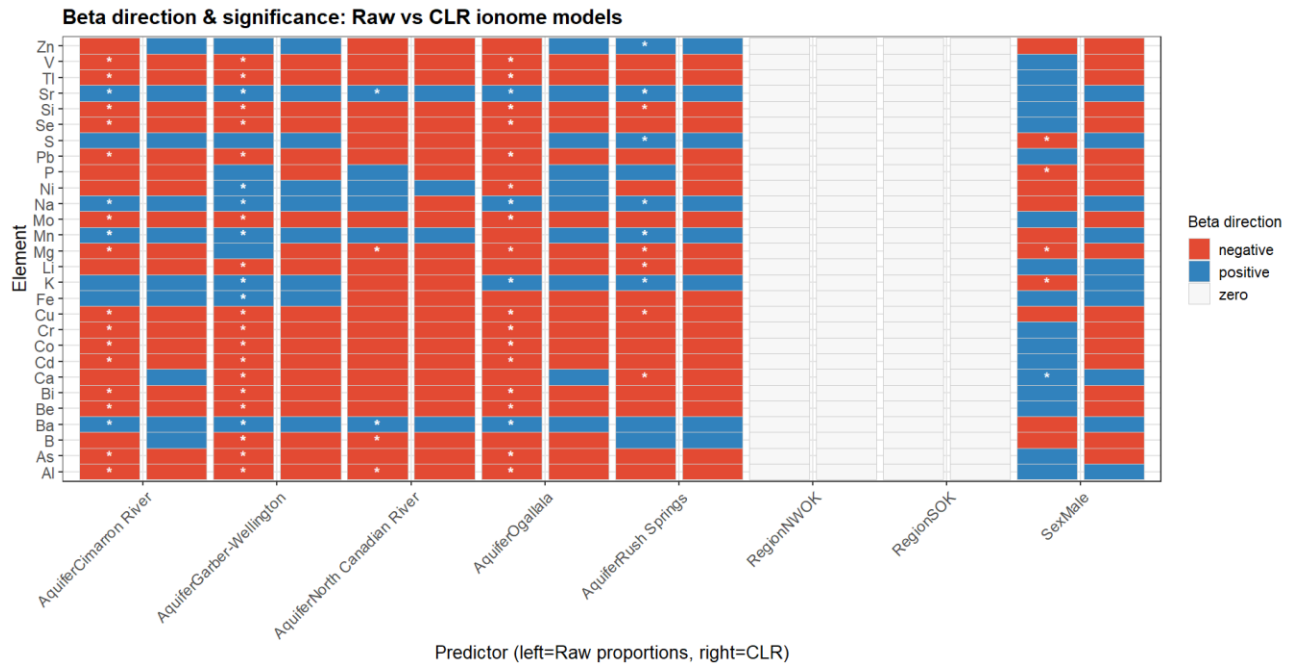
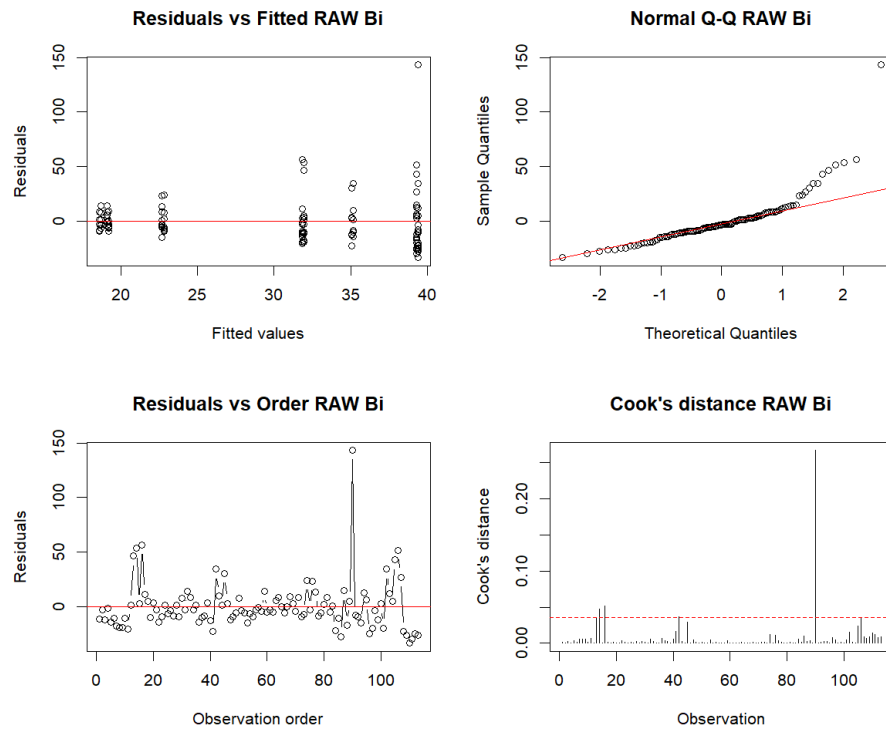


Figure 3.8. Heatmap of coefficient directions under raw and CLR analyses. Heatmap showing the sign and relative magnitude of regression coefficients across predictors and elements under raw (left) and CLR-based (right) models.



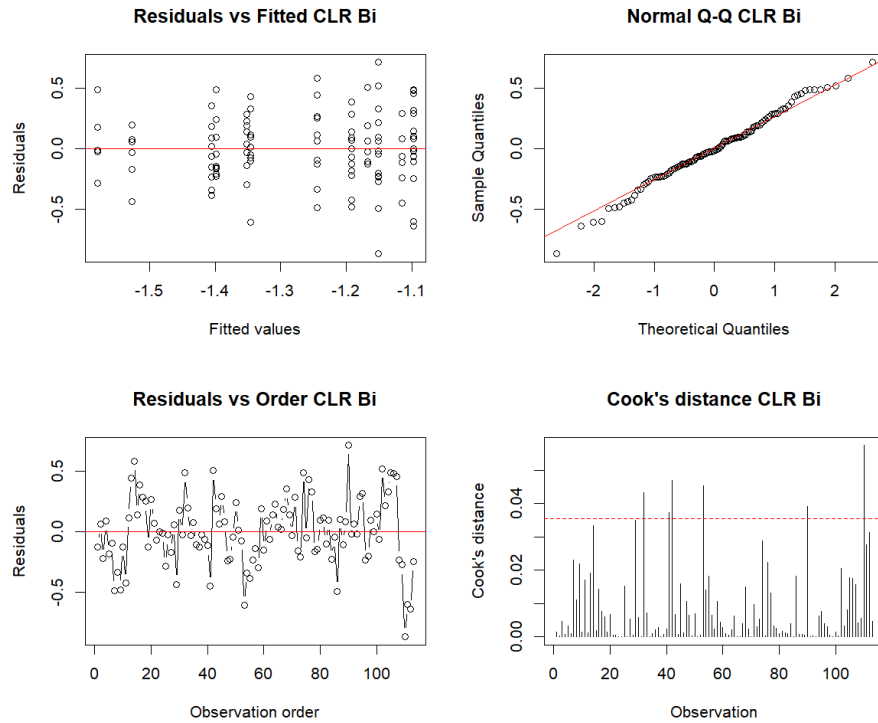


Figure 3.9 and 3.10. Representative diagnostic plots for raw and CLR regressions for the Element Bismuth (Bi). Example fitted-versus-residual, Q–Q, residual-order, and Cook’s distance plots illustrating typical model behavior under raw and CLR representations.

3.5 Summary and Motivation for Simulation

The real-data analysis reveals consistent differences between raw and CLR-based approaches in both visualization and modeling. CLR highlights additional structure in the data, identifies more significant associations, and generally produces better-behaved linear models. At the same time, these improvements are not uniform, and differences in both significance and coefficient direction complicate interpretation.

Because the true underlying relationships in the observed data are unknown, it is not possible to determine directly which method more accurately reflects the biological and environmental processes generating the ionome. These uncertainties motivate a simulation study in which the true data-generating mechanism is known, allowing for direct comparison of methods in terms of Type I error and power. In addition to raw proportions and CLR, a log transformation of the raw proportions is included in the simulation as an intermediate approach motivated by the diagnostic results from the real-data analysis.

4. Simulation Study of Ionomic Data

4.1 Simulation Design

To evaluate the performance of different analytical approaches under controlled conditions, a simulation study was conducted using compositional data generated from a Dirichlet distribution. The simulation design was constructed to mirror the observed ionomic dataset, with the number of components ($D = 28$) and sample size ($n = 113$) matched to the real data. The baseline composition was estimated from the observed data by computing the mean proportion of each element across all samples. Let μ_j denote the baseline mean proportion for element j , with

$$\boldsymbol{\mu} = (\mu_1, \dots, \mu_D) \text{ and } \sum_{j=1}^D \mu_j = 1.$$

For each simulated observation $i = 1, \dots, n$, a predictor X_i was generated independently of the composition. Two predictor types were considered: a binary predictor generated as

$$X_i \sim \text{Bernoulli}(0.5),$$

and a continuous predictor generated as

$$X_i \sim N(0,1).$$

To introduce associations between the predictor and the composition, effects were incorporated on a subset of four affected elements,

$$A = \{1, 2, 3, 4\}.$$

For each element j , the predictor-dependent linear predictor was defined as

$$\eta_{ij} = \log(\mu_j) + X_i \beta_j,$$

where $\beta_j = 0$ for unaffected elements and $\beta_j \neq 0$ for $j \in A$. The corresponding mean composition for observation i was then obtained by exponentiating and renormalizing:

$$\pi_{ij} = \frac{\exp(\eta_{ij})}{\sum_{k=1}^D \exp(\eta_{ik})}, j = 1, \dots, D.$$

Given $\boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{iD})$, the simulated composition $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{iD})$ was generated from a Dirichlet distribution,

$$\mathbf{Y}_i \sim \text{Dirichlet}(\kappa \pi_{i1}, \dots, \kappa \pi_{iD}),$$

where $\kappa = 200$ is a concentration parameter controlling the variability of the compositions around their mean.

Two main scenarios were considered. In the null scenario, no true association was present, so $\beta_j = 0$ for all j . In the effect scenario, nonzero effects were introduced only for the affected set $A = \{1,2,3,4\}$. Within each simulation run, the nonzero effects were assigned random signs and magnitudes, so that the effect structure varied across runs while remaining sparse.

The simulated datasets were analyzed using the same element-wise regression framework described in **Section 3.2.1**. Specifically, for each element j , a linear regression model was fitted under three response representations: raw proportions, log-transformed raw proportions (LogRaw), and centered log-ratio (CLR) values. Reduced and full models were compared using ANOVA to obtain p-values for testing the effect of the predictor.

The simulation study was repeated over $S = 1000$ iterations for each combination of predictor type and scenario. Both unadjusted p-values and Bonferroni-adjusted p-values were recorded. Performance was then summarized using overall Type I error and overall power, as defined in Section 4.2.

4.2 Evaluation Metrics

The performance of each method was evaluated using Type I error and statistical power across S simulation runs. In this study, these quantities were estimated using $S = 1000$ simulation runs. Both Type I error and power were computed using unadjusted and Bonferroni-adjusted p-values.

Under the null scenario, where no true association between the predictor and the compositional response was present for any element, the Type I error for element j was defined as the proportion of simulations in which the null hypothesis for that element was incorrectly rejected:

$$\widehat{\text{Type I Error}}_j = \frac{1}{S} \sum_{s=1}^S I(p_{js} < \alpha),$$

where p_{js} is the p-value for element j in simulation s , α is the significance level, and $I(\cdot)$ is the indicator function. The overall Type I error rate for a given method was then computed as the average of these element-specific rates across all D elements,

$$\text{Overall } \widehat{\text{Type I Error}} = \frac{1}{D} \sum_{j=1}^D \widehat{\text{Type I Error}}_j$$

Under the effect scenario, where a subset of elements was truly associated with the predictor, the power for affected element j was defined as the proportion of simulations in which that element was correctly identified as significant:

$$\widehat{\text{Power}}_j = \frac{1}{S} \sum_{s=1}^S I(p_{js} < \alpha), j \in A,$$

where A denotes the set of truly affected elements. The overall power for a given method was then computed as the average of these element-specific powers across the $m =$ affected elements,

$$\widehat{\text{Overall Power}} = \frac{1}{m} \sum_{j \in A} \widehat{\text{Power}}_j.$$

4.3 Simulation Results

The simulation results are summarized in **Table 4.1** and visually displayed in **Figures 4.1** and **4.2**. Under the null scenario, all three methods adequately controlled Type I error when unadjusted p-values were used. For both binary and continuous predictors, the estimated Type I error rates were close to the nominal level of 0.05. The raw method was slightly conservative in the binary case, while LogRaw and CLR were closer to the nominal level. For example, under the binary predictor with unadjusted testing, the raw method had overall Type I error 0.0324, meaning that, on average across elements, the null hypothesis was incorrectly rejected in about 3.24% of the 1000 simulation runs. When Bonferroni adjustment was applied, Type I error rates decreased substantially for all methods, as expected.

Under the effect scenario, clear differences emerged among the methods. For the binary predictor, raw proportions had relatively low power, whereas both LogRaw and CLR achieved much higher power. A similar pattern appeared for the continuous predictor, where LogRaw and CLR substantially outperformed raw proportions. For example, under the continuous predictor with unadjusted testing, the LogRaw method achieved overall power 0.8125, meaning that, on average, about 81.25% of the affected elements were correctly identified across simulation runs. Although power decreased for all methods under Bonferroni adjustment, the transformed approaches remained markedly more powerful than the raw approach.

Taken together, the simulation results show that the main distinction among methods lies not in their ability to control Type I error, but in their ability to detect true effects. Raw proportions were consistently the least sensitive, while LogRaw and CLR provided substantial gains in power. The similarity between LogRaw and CLR is especially notable, suggesting that in this simulation setting much of the improvement over raw analysis was driven by the use of a log-based transformation rather than by CLR alone.

Table 4.1. Simulated Type I error and power for Raw, LogRaw, and CLR methods across 1,000 simulation runs.

Predictor	Adjustment	Method	Type I error	Power
Binary	Unadjusted	Raw	0.0324	0.1580
Binary	Unadjusted	LogRaw	0.0491	0.4725
Binary	Unadjusted	CLR	0.0486	0.4690
Binary	Bonferroni	Raw	0.0005	0.0148
Binary	Bonferroni	LogRaw	0.0015	0.1915
Binary	Bonferroni	CLR	0.0018	0.1855
Continuous	Unadjusted	Raw	0.0508	0.4515
Continuous	Unadjusted	LogRaw	0.0495	0.8125
Continuous	Unadjusted	CLR	0.0499	0.8095
Continuous	Bonferroni	Raw	0.0014	0.1910
Continuous	Bonferroni	LogRaw	0.0018	0.5933
Continuous	Bonferroni	CLR	0.0016	0.5950

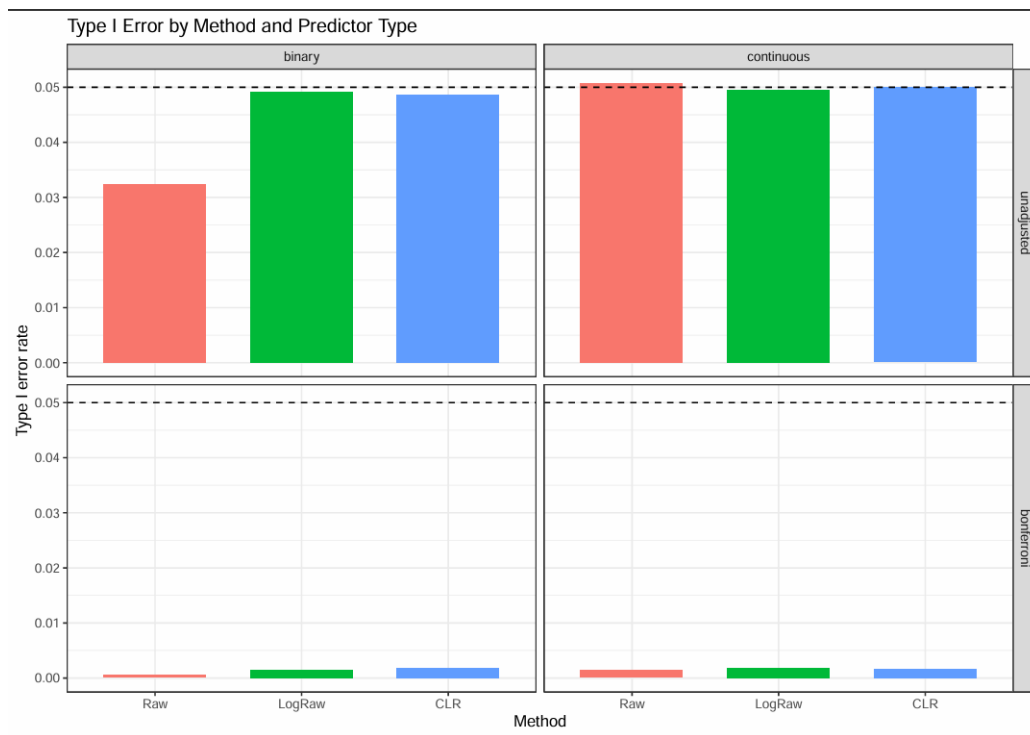


Figure 4.1. Type I error by method and predictor type. Bar plot of simulated Type I error rates for Raw, LogRaw, and CLR methods under binary and continuous predictors, shown for unadjusted and Bonferroni-adjusted analyses.

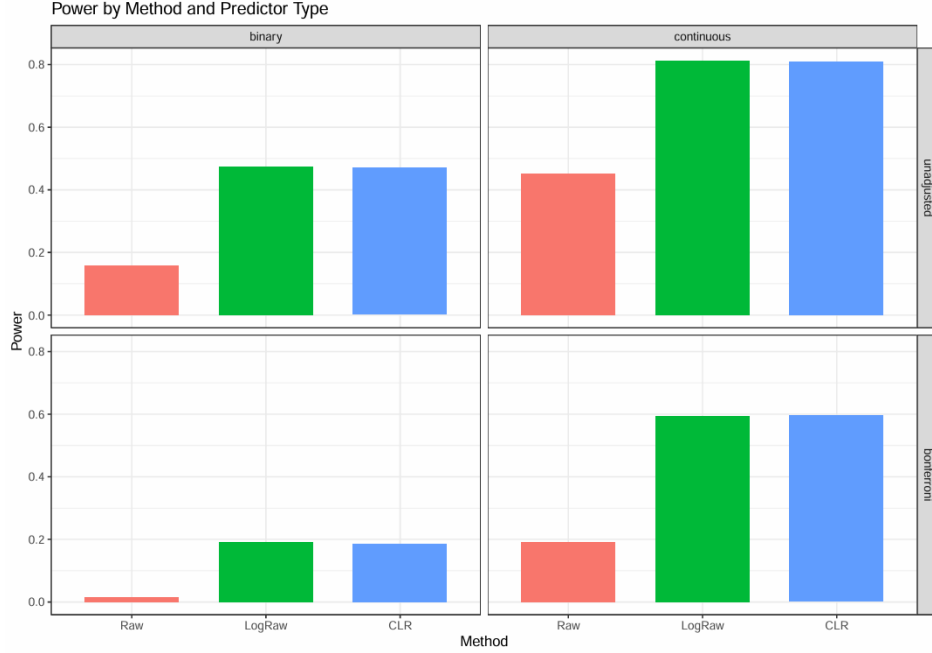


Figure 4.2. Power by method and predictor type. Bar plot of simulated power for Raw, LogRaw, and CLR methods under binary and continuous predictors, shown for unadjusted and Bonferroni-adjusted analyses.

4.4 Beyond Performance: Subcompositional Coherence

In addition to performance measures such as Type I error and power, it is important to consider whether analytical methods satisfy fundamental properties required for coherent inference on compositional data. One such property is subcompositional coherence, which requires that conclusions remain consistent when the analysis is restricted to subsets of components. This property matters in ionomics because different studies or laboratories may measure different sets of elements, exclude some elements, or group elements differently.

To illustrate subcompositional coherence, a separate simulation was conducted with $n = 500$ observations and $D = 25$ components, denoted $C_1, \dots, C_{25}$, where C_j represents the j th part of the composition and $\sum_{j=1}^{25} C_j = 1$. A binary predictor $X_i \sim \text{Bernoulli}(0.5)$ was generated for each observation $i = 1, \dots, n$. The simulated compositions were generated from a Dirichlet distribution with mean

$$\pi_{ij} = \frac{\exp(\alpha_j + \beta_j X_i)}{\sum_{k=1}^D \exp(\alpha_k + \beta_k X_i)}, j = 1, \dots, D,$$

and

$$\mathbf{C}_i = (C_{i1}, \dots, C_{iD}) \sim \text{Dirichlet}(\kappa\pi_{i1}, \dots, \kappa\pi_{iD}),$$

where $\alpha_j = 0$ for all j , $\kappa = 80$, $\beta_1 = 0.5$, $\beta_2 = \dots = \beta_{20} = 3.0$, and $\beta_{21} = \dots = \beta_{25} = 0$. Thus, the true effect was defined on the pairwise log-ratio scale as

$$\beta_1 - \beta_{25} = 0.5$$

for $\log(C_1/C_{25})$.

Analyses were then carried out on the full composition and two reclosed subcompositions: SubA = $\{C_1, C_2, C_3, C_4, C_5, C_{25}\}$, which contains several strongly affected components, and SubB = $\{C_1, C_{21}, C_{22}, C_{23}, C_{24}, C_{25}\}$, which contains mainly weakly affected or unaffected components. For each compositional context S , three analysis targets were considered: the raw proportion of C_1 ,

$$R_{i1}^{(S)} = \frac{C_{i1}}{\sum_{j \in S} C_{ij}},$$

the CLR of C_1 ,

$$Z_{i1}^{(S)} = \log \left(\frac{R_{i1}^{(S)}}{g(R_i^{(S)})} \right),$$

and the pairwise log-ratio

$$L_i = \log \left(\frac{C_{i1}}{C_{i25}} \right).$$

Here, $g(R_i^{(S)})$ denotes the geometric mean of the reclosed subcomposition for observation i . Each target was regressed on X_i using a simple linear model:

$$\begin{aligned} R_{i1}^{(S)} &= \beta_0^{(R,S)} + \beta_1^{(R,S)} X_i + \varepsilon_i^{(R,S)}, \\ Z_{i1}^{(S)} &= \beta_0^{(C,S)} + \beta_1^{(C,S)} X_i + \varepsilon_i^{(C,S)}, \\ L_i &= \beta_0^{(P)} + \beta_1^{(P)} X_i + \varepsilon_i^{(P)}. \end{aligned}$$

In these models, the slope coefficients $\beta_1^{(R,S)}$, $\beta_1^{(C,S)}$, and $\beta_1^{(P)}$ represent the effect of the binary predictor X_i on the chosen representation of C_1 . Specifically, $\beta_1^{(R,S)}$ is the expected change in the raw proportion of C_1 within compositional context S when X_i changes from 0 to 1. Likewise,

$\beta_1^{(C,S)}$ is the expected change in the CLR-transformed value of C_1 within context S , and $\beta_1^{(P)}$ is the expected change in the pairwise log-ratio $\log\left(\frac{C_1}{C_{25}}\right)$ when X_i changes from 0 to 1.

The resulting estimates are shown in **Table 4.2** and displayed in **Figure 4.3**. For the raw analysis, the estimated slope in the full composition was -0.036 , meaning that when X_i changes from 0 to 1, the expected raw proportion of C_1 decreases by about 0.036 on average when all 25 components are included. In SubB, however, the raw slope became $+0.087$, meaning that under the reduced composition the same predictor is associated with an increase of about 0.087 in the expected reclosed proportion of C_1 . This sign reversal shows that the apparent effect of X_i on C_1 depends on which other components are included in the composition.

For the CLR analysis, the estimated slope changed from -2.86 in the full composition to $+1.84$ in SubB. Thus, the estimated effect of X_i on the relative abundance of C_1 , measured against the geometric mean of the included components, also changes direction when the component set changes. By contrast, for the pairwise log-ratio analysis, the estimated slope was $+1.97$ in the full composition, SubA, and SubB. This means that when X_i changes from 0 to 1, the expected value of $\log(C_1/C_{25})$ increases by about 1.97 units on average, regardless of which additional components are present. This invariance is the key evidence of subcompositional coherence.

Table 4.2. Estimated effects under the full composition and two subcompositions. Consistency across columns indicates subcompositional coherence.

Method	Full	SubA	SubB
Raw	-0.036	-0.142	+0.087
CLR	-2.86	-2.53	+1.84
PLR	+1.97	+1.97	+1.97

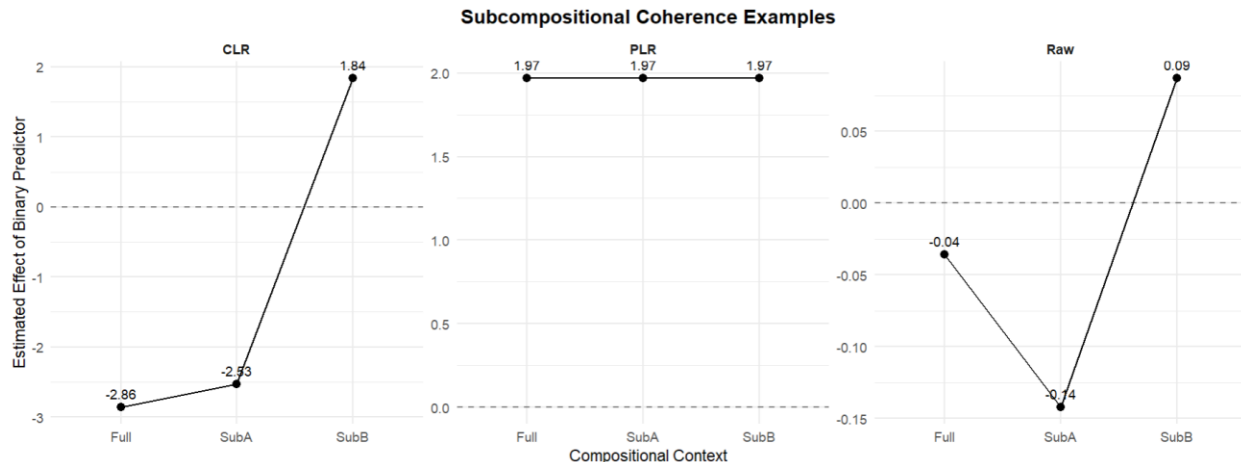


Figure 4.3. Graphical display of the subcompositional coherence results: Three-panel plot showing the estimated effect of the binary predictor for the full composition and two subcompositions (SubA and SubB) under raw, CLR, and pairwise log-ratio analyses. The figure highlights the instability of raw and CLR estimates across subcompositions and the invariance of the pairwise log-ratio estimate.

5. Discussion and Conclusion

5.1 Main Findings and Interpretation

The results of this study show that the way ionomic data are represented has a meaningful effect on both statistical performance and scientific interpretation. In the real-data analysis, raw proportions and CLR-based representations did not simply provide alternative views of the same structure; they produced different visual patterns, different model behavior, and different inferred relationships among elements. Raw proportions emphasized dominant components and broad compositional structure but often compressed smaller elements and obscured subtler variation. In contrast, CLR revealed additional structure and produced more stable model behavior across many elements.

The simulation study clarified these real-data patterns under known data-generating conditions. All three approaches controlled Type I error adequately, but raw proportions consistently had the lowest power, while both LogRaw and CLR performed substantially better. Taken together, the real-data and simulation results suggest that some log-based transformation appears necessary in ionomics if the goal is to conduct reliable statistical inference.

At the same time, the similarity between LogRaw and CLR is one of the most important findings of this study. It suggests that much of the improvement over raw analysis may come from applying

a log-based transformation that reduces skewness and stabilizes variance, rather than from CLR alone. In practical terms, a log transformation of raw proportions may already provide substantial improvement over untransformed analysis in many ionomic settings.

However, comparable performance in power does not imply equivalence in interpretation. LogRaw is primarily a pragmatic transformation that improves modeling behavior, whereas CLR is designed specifically for compositions and expresses each part relative to the geometric mean of the whole. These two methods become more similar with smaller variability in the geometric means across observations, but they can be quite different when this variability is high. This distinction matters because ionomic data are interpreted in relative terms: increasing one element necessarily changes the others. LogRaw can improve model fit and detection, but it does not explicitly account for this dependence among components. CLR, by centering each element against the geometric mean of the full composition, provides estimates that are more directly interpretable as changes in relative composition.

These findings lead to a balanced but clear conclusion. Raw proportions are generally not adequate for formal ionomic inference. A log-based transformation appears necessary. Whether CLR is required depends more on the researcher's inferential goal than on power alone.

5.2 Subcompositional Coherence and Its Implications

Subcompositional coherence adds an important dimension to the interpretation of these results. In ionomics, researchers may not always measure the same set of elements. Different laboratories may focus on different elemental panels; some studies may exclude low-abundance elements, and others may group elements differently depending on the scientific question. In such settings, it is desirable that conclusions remain stable when the analysis is restricted to a subset of components.

The coherence demonstration showed that neither raw proportions nor CLR produced stable estimates across subcompositions. The estimated effect for the same component changed when the analysis moved from the full composition to reduced subsets, sometimes even reversing signs. This occurs because both methods depend on the full set of included components: raw proportions change through reclosure, while CLR changes through the recomputed geometric mean.

By contrast, the pairwise log-ratio (PLR) approach produced identical results across the full composition and both subcompositions. When the inferential target is defined as a ratio between components, the estimate remains stable regardless of which other parts are present. More generally, coherent log-ratio approaches such as PLR, ALR, and ILR have an advantage that is not captured by power and Type I error alone.

5.3 Connection to Literature

The findings of this study fit naturally within the broader debate on the practical importance of compositional data analysis. Prior work in microbiome and metabolomics research has shown that raw relative abundances can be misleading and that log-ratio methods often improve validity and interpretation. The present results are consistent with that literature in showing that raw proportions are generally the weakest option, both in terms of model behavior and statistical power.

At the same time, the results also support more recent arguments that the practical gains of compositional methods depend on context. In particular, the strong empirical performance of LogRaw is consistent with the view that simpler transformations may recover much of the benefit associated with formal CoDA approaches in some settings. This is especially relevant when the goal is screening, exploratory modeling, or improved detection rather than explicitly compositional interpretation.

The study also aligns with the caution that CLR is not always the most interpretable or practically necessary choice. However, the subcompositional coherence results add an important qualification: even when CLR does not clearly outperform LogRaw in power, methods that do not preserve coherence may still be limited for certain inferential goals. This suggests that the question is not simply whether CLR improves performance, but which method is most appropriate for the kind of conclusion the researcher wants to draw.

5.4 Practical Recommendations for Ionomics

Several practical recommendations follow from this study. First, raw proportions should generally not be used directly for formal inference in ionomics. They tend to yield lower power, poorer model behavior, and interpretations that are strongly influenced by dominant elements.

Second, some log-based transformation appears necessary. Both LogRaw and CLR substantially improved performance relative to raw proportions in the simulation study, and this was consistent with the improved diagnostics and clearer structure observed in the real-data analysis. Researchers working with ionomic data should therefore treat transformation as a basic requirement rather than an optional refinement.

Third, the choice between LogRaw and CLR should depend on the inferential goal. If the main aim is to improve linear modeling and increase sensitivity to detect associations, LogRaw may serve as a simple and effective option. If the goal is to interpret variation explicitly in relative compositional terms, CLR is more appropriate because it directly accounts for the constant-sum constraint.

Finally, if subcompositional coherence matters, then PLR methods should be preferred. This is especially relevant when comparing studies that measure different sets of elements, when analyzing subsets of elements, or when the scientific question centers on interpretable relationships between specific components. In such cases, coherent log-ratio methods provide the most defensible framework.

5.5 Limitations of this Study

This study conducts an initial exploration to understand the effect of using different CoDA approaches for ionomic data, but it has several limitations. The main simulation framework was based on a Dirichlet model with log-linear effects, which may favor log-based methods and may not represent all forms of ionomic variation encountered in practice. Different conclusions might arise under alternative data-generating mechanisms, including settings where relationships are closer to linear on the raw scale.

The real-data analysis was based on one biological system, namely amphipod samples collected from freshwater springs in Oklahoma. Although this provides a realistic applied context, the findings may not generalize automatically to all ionomic studies, especially those involving different organisms, tissues, or sampling designs.

The modeling framework was intentionally simple. Linear models were used throughout, and the main comparisons focused on raw, LogRaw, and CLR representations in simulation and on raw and CLR in the real data. Other approaches, including ILR coordinates, balances, multivariate compositional models, and methods tailored to high-dimensional compositions, were not explored.

In addition, the real-data comparison relied on unadjusted p-values and was interpreted in a comparative rather than a confirmatory sense. Although multiple testing was incorporated in the simulation study, stronger control procedures could be emphasized in future applied analyses. Finally, while subcompositional coherence was examined through a targeted simulation, coherent methods such as PLR or ILR were not incorporated into the full real-data workflow.

5.6 Future Work

Most of the above limitations arise due to the simplicity of the study due to time constraints. Several directions for future work follow naturally from these results. A first priority is to extend the analysis to ILR-based and balance-based methods in ionomics, particularly because they combine compositional validity with subcompositional coherence. Such methods may offer a better compromise between interpretability and statistical rigor than CLR in some settings.

A second direction is to broaden the simulation framework. Future simulations could consider alternative data-generating processes, including non-log-linear mechanisms, different sparsity patterns, stronger dependence structures, and varying numbers of measured elements. This would help determine whether the similarity between LogRaw and CLR persists more generally.

A third direction is to evaluate these methods across a wider range of ionomic datasets and biological systems. Doing so would clarify how robust the present conclusions are to differences in organism, environment, and measurement design. Future work could also investigate coherence and comparability more directly in multi-study settings, where different laboratories measure different subsets of elements and subcompositional consistency becomes especially important.

Finally, this study considered only the case in which ionomic compositions served as response variables. Future research should extend this framework to settings where compositions are used as predictors, since the statistical and interpretational implications may differ in important ways.

5.7 Conclusion

This study examined the practical importance of compositional methods in ionomics through a combination of real-data analysis, simulation, and a focused evaluation of subcompositional coherence. The results show that raw proportions are generally the weakest option for inference: they obscure smaller components, produce less stable models, and have substantially lower power than transformed alternatives.

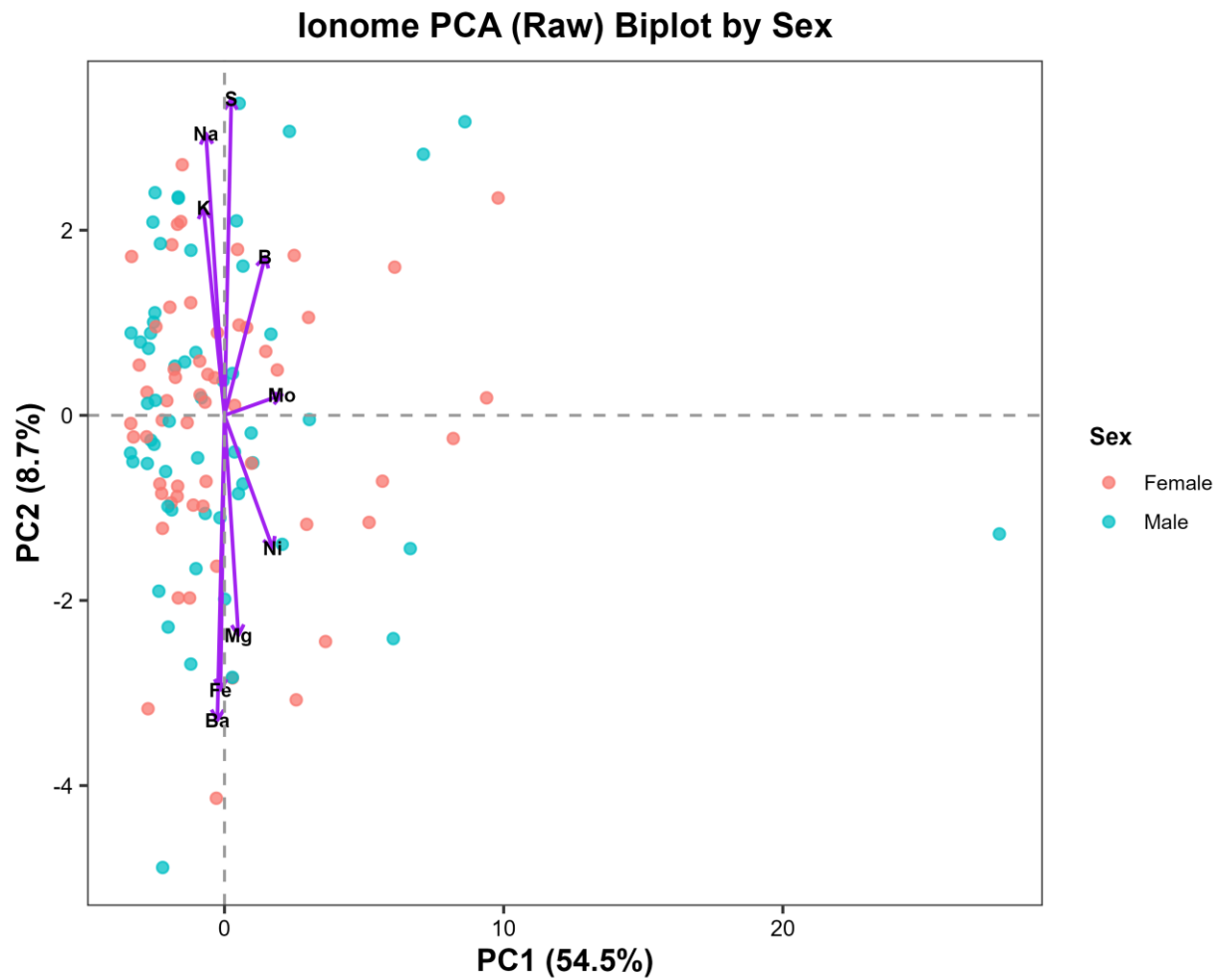
At the same time, the findings do not support an overly simple conclusion that CLR is always necessary in practice. In the main simulation study, LogRaw performed nearly as well as CLR while maintaining Type I error control, suggesting that some log-based transformation appears necessary, but CLR may not always be uniquely required for detection. However, the coherence analysis showed that when the scientific question requires conclusions to remain stable across subsets of elements, raw, LogRaw, and CLR are not sufficient. In that setting, pairwise or isometric log-ratio approaches provide the strongest framework.

Overall, the main message of this study is that method choice in ionomics should be guided by inferential goals. For general modeling and signal detection, a log-based transformation appears necessary. For explicitly compositional interpretation, CLR remains preferable to raw or LogRaw. And for coherent inference across subcompositions, PLR or ILR-type methods are the most defensible choice.

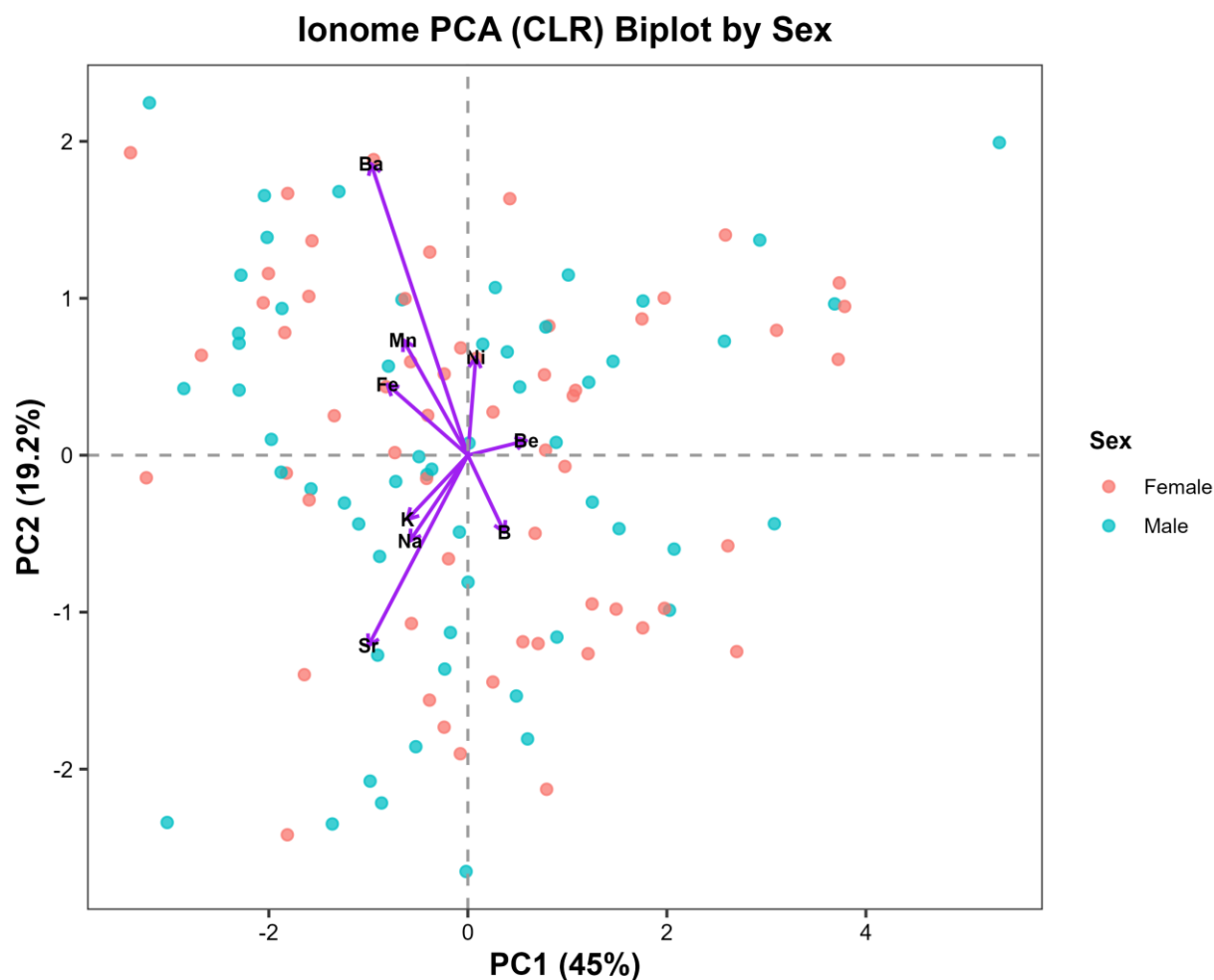
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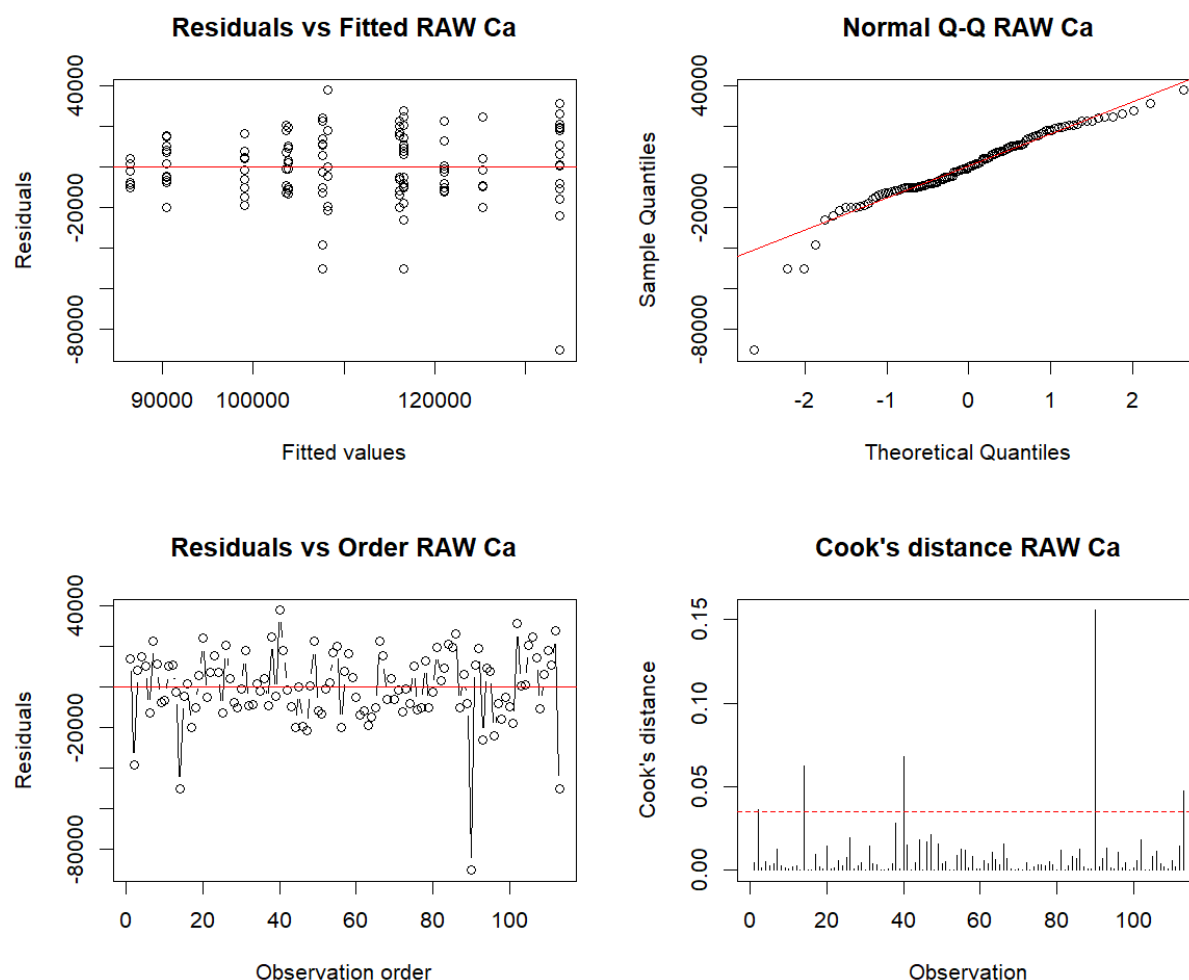
Appendix



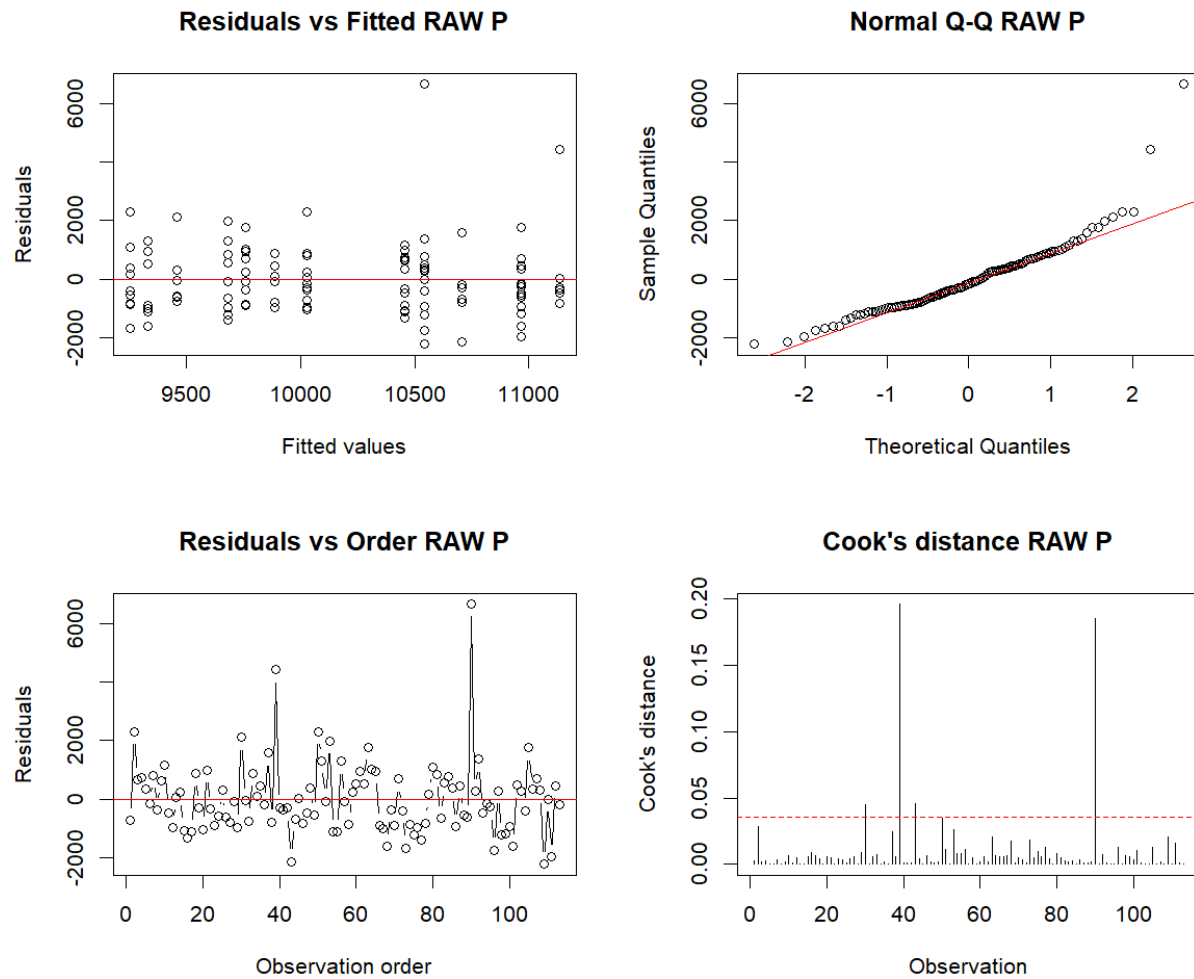
Appendix Figure A.1. Principal component analysis of raw ionomic data by sex. Biplot of the first two principal components from PCA applied to centered and scaled raw elemental data, with points colored by sex. The raw representation concentrates a large share of the variation in the first principal component and shows substantial overlap between female and male samples. For clarity, only a subset of element loadings is displayed, namely those with contributions to the first two principal components at or above the median contribution.



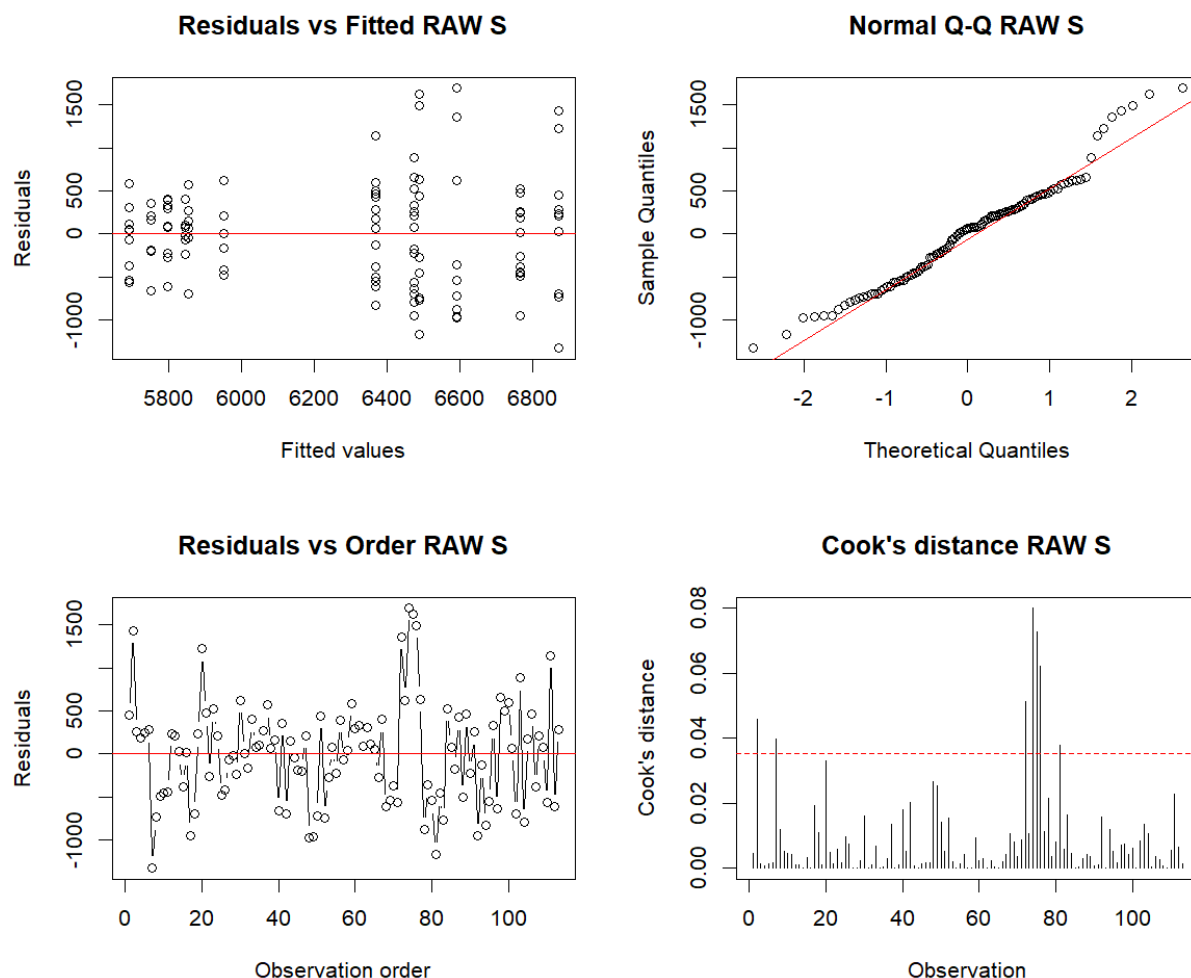
Appendix Figure A.2. Principal component analysis of CLR data by sex. Biplot of the first two principal components from PCA applied to centered CLR-transformed elemental data, with points colored by sex. Compared with the raw PCA, variation is distributed more evenly across the first two principal components, although female and male samples still show substantial overlap. For clarity, only a subset of element loadings is displayed, namely those with contributions to the first two principal components at or above the median contribution.



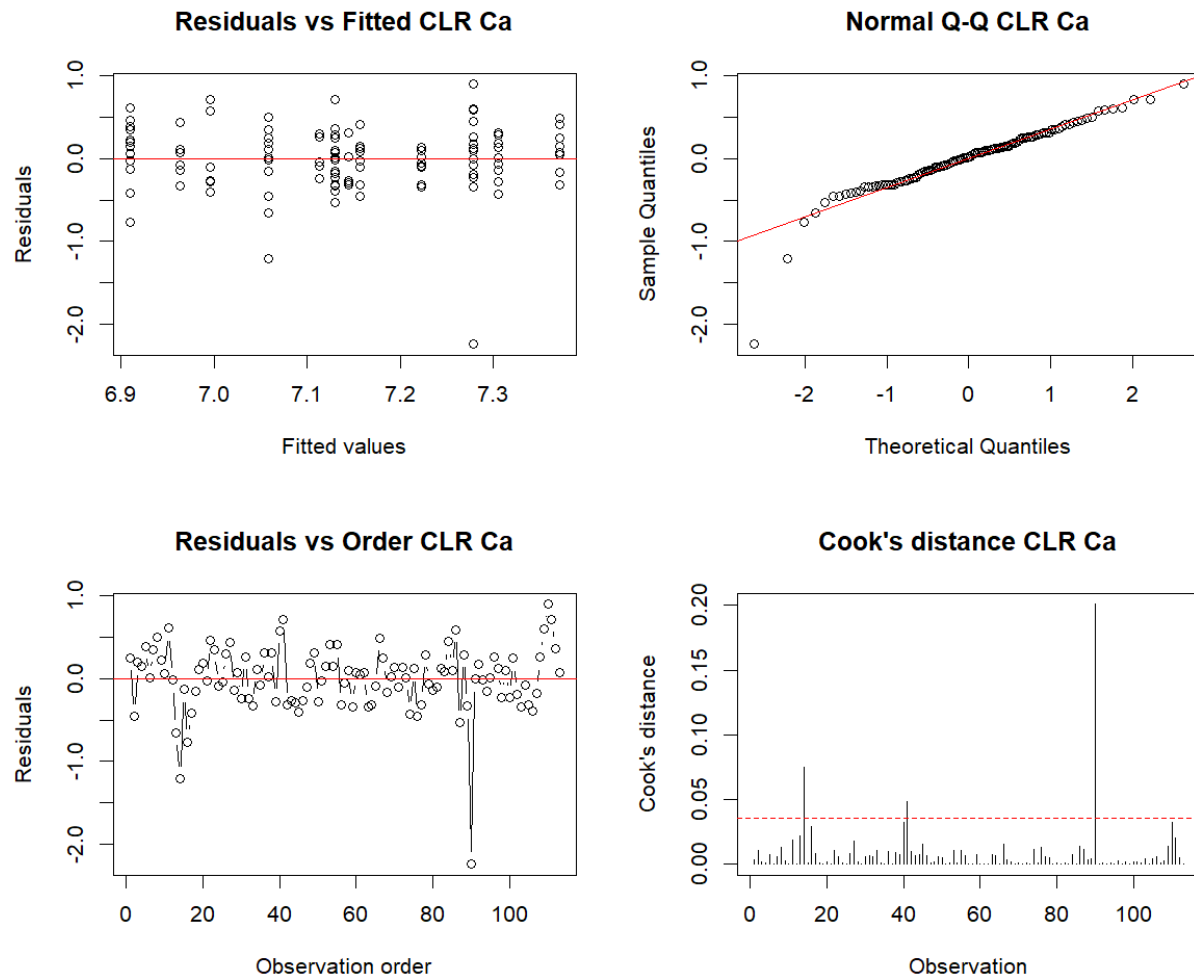
Appendix Figure A.3. Diagnostic plots for the raw regression model for calcium (Ca). *Residuals-versus-fitted, normal Q-Q, residuals-versus-order, and Cook's distance plots for the raw proportion model for Ca. The diagnostics suggest departures from normality, substantial residual spread, and the presence of influential observations, illustrating the modeling challenges associated with highly abundant elements on the raw scale.*



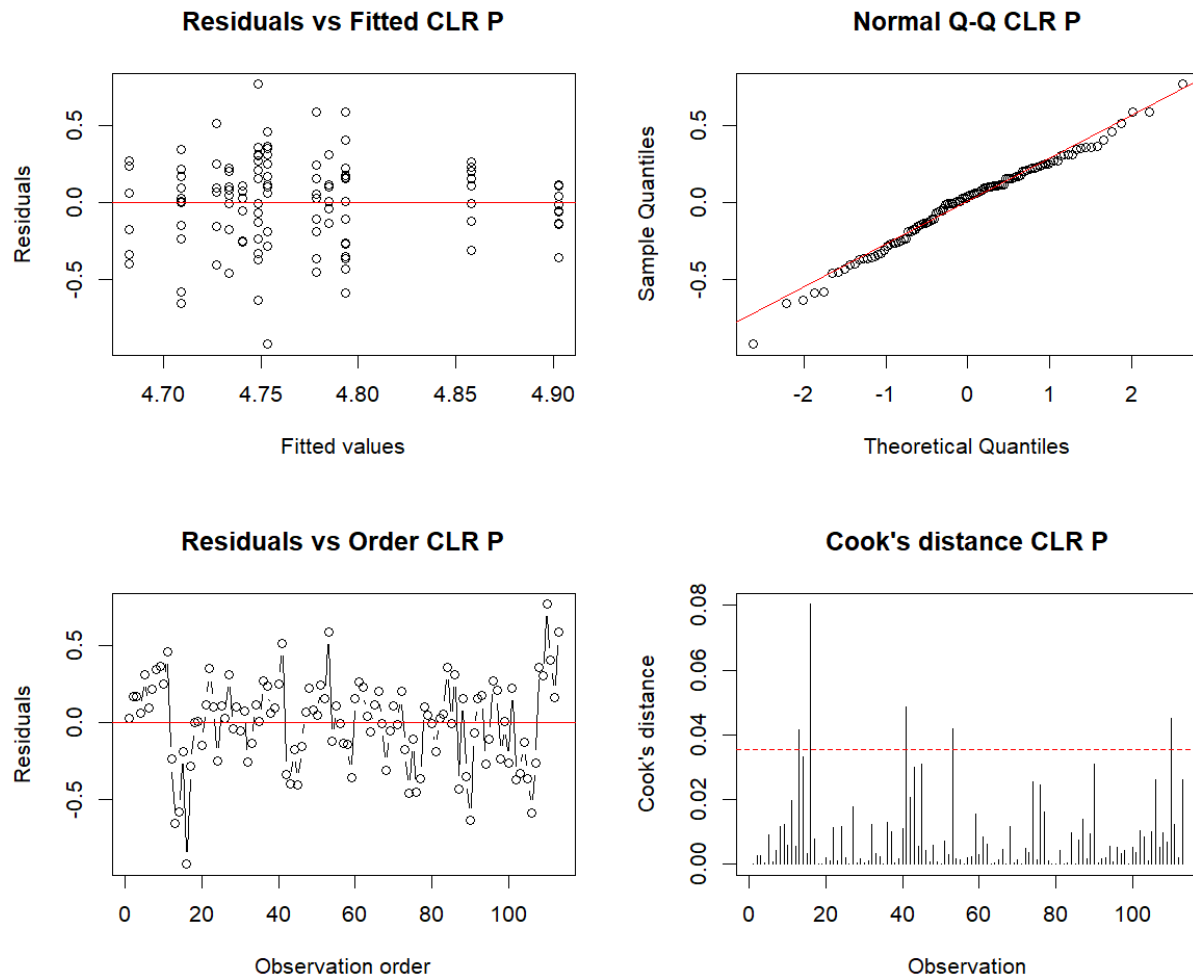
Appendix Figure A.4. Diagnostic plots for the raw regression model for phosphorus (P). *Residuals-versus-fitted, normal Q-Q, residuals-versus-order, and Cook's distance plots for the raw proportion model for P. The plots indicate tail departures from normality and several influential observations, although the residual spread is less extreme than for Ca.*



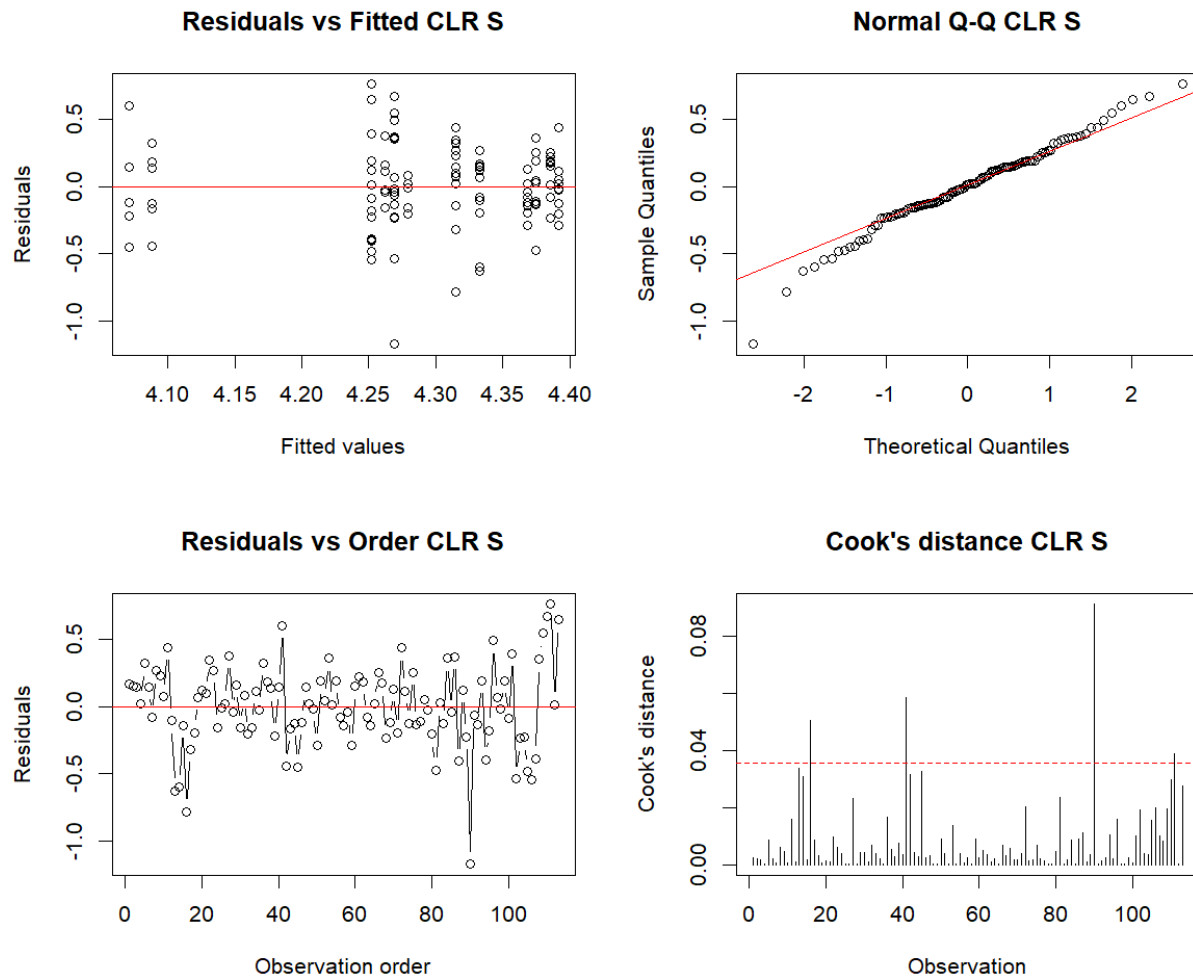
Appendix Figure A.5. Diagnostic plots for the raw regression model for sulfur (S). *Residuals-versus-fitted, normal Q-Q, residuals-versus-order, and Cook's distance plots for the raw proportion model for S. The diagnostics show moderate deviations from model assumptions, including non-normality in the tails and several observations with elevated influence.*



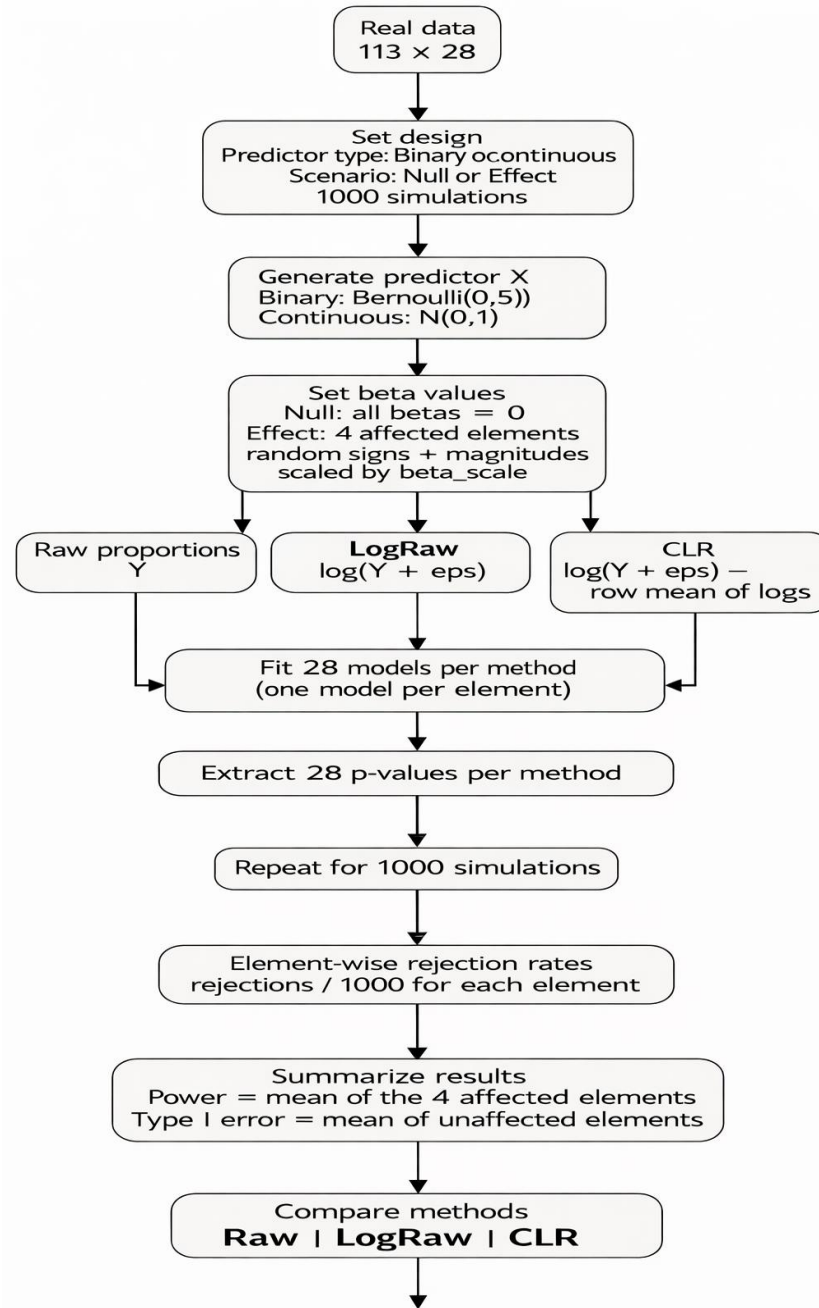
Appendix Figure A.6. Diagnostic plots for the CLR regression model for calcium (Ca). *Residuals-versus-fitted, normal Q-Q, residuals-versus-order, and Cook's distance plots for the CLR-transformed model for Ca. Compared with the corresponding raw model, the residuals are more tightly concentrated and the Q-Q plot is closer to linear, although mild tail departures and one highly influential observation remain.*



Appendix Figure A.7. Diagnostic plots for the CLR regression model for phosphorus (P). *Residuals-versus-fitted, normal Q-Q, residuals-versus-order, and Cook's distance plots for the CLR-transformed model for P. The diagnostics suggest more stable residual behavior than in the raw model, with improved normality and more moderate spread, though several observations still show elevated influence.*



Appendix Figure A.8. Diagnostic plots for the CLR regression model for sulfur (S). *Residuals-versus-fitted, normal Q-Q, residuals-versus-order, and Cook's distance plots for the CLR-transformed model for S. The plots indicate generally improved residual behavior relative to the raw model, with tighter spread and closer agreement with normality, although mild tail departures and a small number of influential observations persist.*



Appendix Figure A.9. Flowchart of the main simulation design.

Overview of the simulation workflow used to compare Raw, LogRaw, and CLR approaches. The design begins with the real ionomic dataset to match sample size and dimensionality, then generates either a binary or continuous predictor under null and effect scenarios. For each simulation, element-specific models are fitted under all three methods, p-values are extracted, and element-wise rejection rates are summarized across 1,000 runs to obtain overall power and Type I error measures.